1 Excellence

1.1 Objectives

The overall objective of the PROLIFIC project is to apply, validate and scale up an integrated array of PROcesses, to recover a significant amount of proteins/peptides and other value-added compounds (e.g. carotenoids, phenols and fibres) from agro-industrial processing residues of legumes (seeds of peas, beans and chickpeas), fungi (cuttings and mycelia of different species) and coffee (silver skin and not compliant roasted seeds). The recovery of these ingredients will be performed applying cascades of processes enabling the sequential extraction of valuables in the following order: protein, fibres, bioactive ingredients and lastly residues. A wide range of LCA- and CEA-proofed economically and environmentally sustainable extraction, modification and stabilization techniques will exclusively be used/developed "with and for" end-users from four industrial sectors. The resulting valuable molecules and fractions are high-value ingredients for the formulations of food, cosmetic, packaging and feed product prototypes fully complying with current safety and regulation standards.

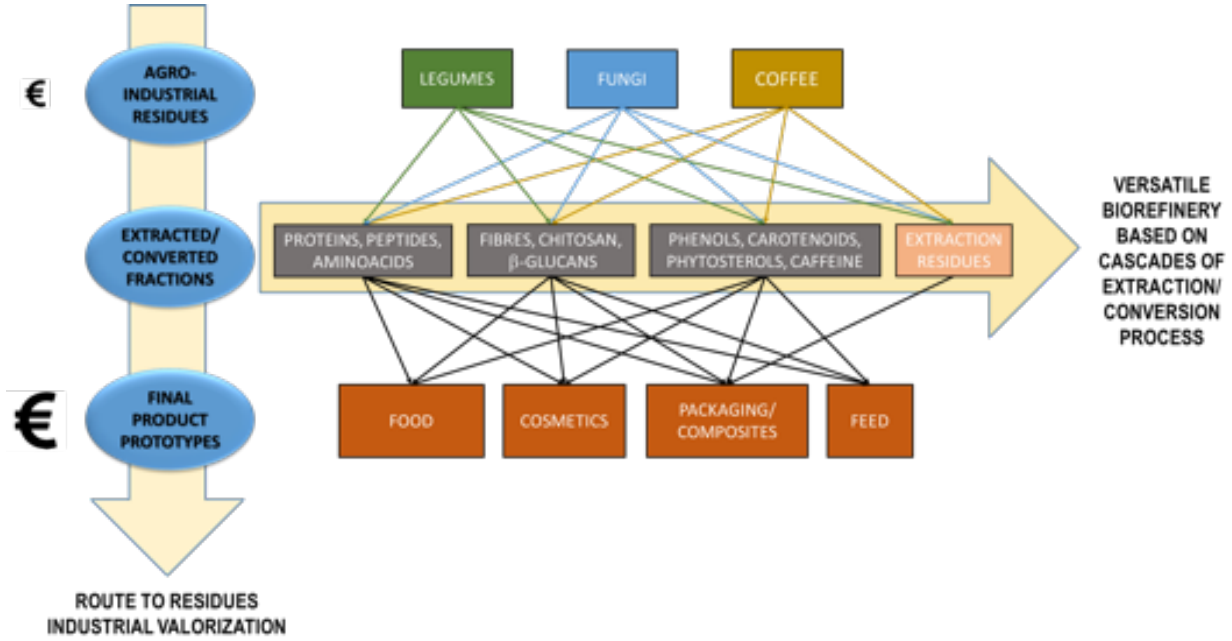

Figure 1. PROLIFIC routes legume, fungi and coffee agro-industrial residues to industrial valorisation through formulations based on extracted fractions and realisation of product prototypes.

The PROLIFIC project sets its activity in the frame of the Circular Economy Package[1] launched in 2017 by DG Environment of the European Commission, that strongly promotes the adoption of cascades of processes to use biomass in integrated biorefineries and the innovative use of co-products, side-products, residual streams and bio-waste to obtain value-added bio-based chemicals and materials to address specific market demands.

Among possible biomass, those deriving from the entire food chain (from the farmers to the consumers) are by far those generating yearly the largest amount of co-products, residues and wastes (1.3 BT/year globally; FAO[2]). Those residues are yet a rich source of useful compounds like carbohydrates, proteins, lipids and polyphenols, which could find high-value applications in many industrial sectors. Most importantly, innovative and sustainable biorefinery processes starting from protein-rich side products could significantly contribute to face the projected global demand for food proteins (largely exceeding current production capacities) which is expected to increase by 76% in 2050[3] due to the steadily growing world population. This predicted shortage in proteins resource is amplified by the increasing income *per capita* in industrialising countries in Asia and South and Middle America[2]. In particular, the increase in the demand for animal protein will result in a tremendous growth in the need for feed protein[4], which will be hard to meet in an eco-sustainable way given the high resource footprint of producing animal-derived food products (projected 80% increase in greenhouse gases by 2050[5]). Possible technological option to enhance the sustainability of future protein supply is to increase the use of plant proteins for protein ingredients, which was around

[1] http://ec.europa.eu/environment/circular-economy/index_en.htm
[2] http://www.fao.org/save-food/resources/keyfindings/en/
[3] DG Research and Innovation, European Research and Innovation for Food and Nutrition Security, Food 2030 Conference document
[4] https://ec.europa.eu/agriculture/sites/agriculture/files/markets-and-prices/medium-term-outlook/2016/2016-fullrep_en.pdf
[5] Growing Greenhouse Gas Emissions Due to Meat Production, UNEP 2012

4.5 MT in 2015 and is expected to reach over 8.0 MT by 2025, growing at a Compound Average Growth Rate (CAGR) of 6.2% from 2016 to 2025. In terms of value the global protein ingredients market size was valued at 23.98 billion USD in 2015 and is expected to witness growth at a CAGR of over 7% from 2016 to 2024[6].
Residual biomass and side streams are also potentially interesting sources of other bioactive compounds among which some of the most important for their industrial applications are fibres, polyphenols, carotenoids and phytosterols and caffeine. Fibres are recognized as an important part of healthy diet and lifestyle as their consumption helps to prevent several diseases, such as diabetes, several types of cancer and cardiovascular pathologies. The global market for high fibre content food has been estimated to reach almost 30 billion USD by 2020 and is intensively driven by the growing consumer focus on health, wellbeing and increasing awareness on the benefits of fibre-rich diet[7]. Many discarded agro-industrial and food processing leftovers are indeed rich in soluble and insoluble fibres (about 3.7 MT of total vegetal waste in 2014[8]), which might be reintroduced into the food chain and can also find applications in other markets (such as cosmetics, plastics and feeds). Polyphenols are moderately water-soluble plant metabolites. They have antioxidant, anti-inflammatory, antibacterial and antiviral properties and are used in numerous sectors of the food-processing industry as natural colouring, conservative, antioxidants and nutritional additives. In addition, many polyphenol-rich plant extracts are actually integrated into cosmetic or pharmaceutical formulations. The global polyphenols market size was 16.54 kT in 2015 and is expected to reach 33.88 kT by 2024, growing at a CAGR of 8.4% from 2016 to 2024[9]. The key drivers impacting the polyphenols market are favourable food safety regulations and increasing consumer awareness of health benefits associated with their consumption. The leading companies in polyphenol extracts market are involved in new developments and innovations to increase the application areas of polyphenols across various industries. Carotenoids (*e.g.* α-carotene, lycopene, lutein) are a widespread group of fat-soluble pigments and represent one of the major food micronutrients. In developed countries, 80–90% of the carotenoid intake comes from fruit and vegetable and data evidenced that 50 different carotenoids can be absorbed and metabolized by the human body[10]. Carotenoids play an important role in human health by acting as biological antioxidants and as precursors of Vitamin A. A deficiency of the latter is a major cause of premature death in developing nations, particularly among children[11],[12]. Extensive data regarding the most commonly consumed carotenoid-rich foods in USA and EU are reported in The Carotenoid Content of USA Foods' and in the European databases[13],[14]. Phytosterols are ubiquitous in plants, in particular in Leguminoseae and Brassicaceae. They are assimilated through the daily diet intake and may have beneficial effects on humans such as the prevention of breast and prostate cancer, cardiovascular disease and osteoporosis[15]. The European Food Safety Authority (EFSA) approved the addition of phytosterol extracts in various foods (*i.e.,* soy drinks, salad dressings, and rye bread) at levels resulting in intake of up to 3 g/day[16]. Phytosterols were also assessed for cosmetic use[17] and reported to function as skin and hair conditioning agents, viscosity increasing agents, skin protectants, antioxidants, and fragrances. Phytosterols are used in all personal care products categories at concentrations ranging from 0.000001% - 8%. Coffee is the primary source of caffeine. Caffeine may have both positive (antioxidant, protective effect against Parkinson's disease and osteoporosis, *in vitro* growth inhibition of cancer cells) and negative effects (insomnia, increased blood pressure and heart rate, irritability)[18]. Caffeine is classified by the USA FDA[19] as generally safe as toxic doses, over 10 grams per day for an adult, are much higher than typical daily doses (80–175 mg of caffeine per cup). Caffeine is used as additive in soft drinks, drug preparations and cosmetic formulations.

PROLIFIC will take on the resource challenge by developing sustainable technologies and procedures to recover proteins, fibres and other bioactive compounds from processing residues and valorise them in the formulation of different product prototypes such as cereal-based, vegetarian/vegan and meat food products, cosmetic creams and toothpaste, bioactive and biodegradable food and cosmetic packaging (i.e. bioactive and biodegradable coffee capsules), feed for pigs and poultry. Protein and derivates thereof will be sourced from legumes, fungi and coffee, and fibres from legumes and coffee residues. The targeted bioactive compounds are chitosan and β-glucans (from

[6] Grand View Research 2016- Protein Ingredients Market Analysis By Product [Plant (Wheat, Soy Concentrates), Animal (Egg, Milk Concentrates)] By Application (Food & Beverages, Infant Formulations, Personal Care & Cosmetics, Animal Feed), And Segment Forecasts, 2014 - 2025
[7] Global Industry Analysts, Inc., 2014
[8] http://ec.europa.eu/eurostat/web/main
[9] https://www.reportlinker.com/p04144317/Polyphenols-Market-Analysis-By-Product-By-Application-And-Segment-Forecasts-To.html
[10] Khachik, F et al. (1997). Anal. Chem. 1997, 69, 1873–1881.
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[14] .Holden J.-M. et al. (1999). J. Food Comp. Anal., 12, 169 –196.
[15] Mastrodi Salgado J, Donado-Pestana C (2011). In: El-Shemy H (ed) Soybean and Nutrition, InTech, Shangai, China pp 21-44
[16] European Food Safety Authority (EFSA). Opinion of the Scientific Panel on Dietetic Products, Nutrition and Allergies on a request from the Commission related to a notification from Cognis, ADM and Cargill on vegetable oils-derived phytosterols and phytosterol esters from soybean sources pursuant to Article 6 paragraph 11 of Directive 2000/13/EC. EFSA. 2007. http://www.efsa.europa.eu/fr/efsajournal/pub/486.htm. Report No. EFSA-Q-2006-162. pp. 1-8.
[17] Safety assessment of phytosterols as used in cosmetic. Cosmetic Ingredient Review, 2013
[18] Cano-Marquina A. et al. (2013). Maturitas, 75, 7-21
[19] https://www.fda.gov

fungi), polyphenols (from legumes, fungi and coffee), carotenoids and phytosterols (from legumes) and caffeine (from coffee).

The project will develop/apply versatile biorefinery processes for three different biomass streams. These processes will be used in cascades to recover ingredients to be used for the formulation of new product prototypes (Fig. 1). The goal is to upcycle significant amounts of the selected agro-industrial residues that are currently disposed/discarded or have low value use (*e.g.* composting). Our choice of residues is based on a preliminary assessment, which shows that these feedstocks are available in significant amounts. This will enable the further upscaling of the developed extraction/modification processes to produce quantities of ingredients satisfying the needs of our industrial end-users for the production of final value-added product prototypes in the food, cosmetic, packaging and feed sectors. Measures necessary (incl. safety and regulatory aspects) to prepare market entry of the extracted ingredients and product prototypes will be taken. To turn this ambitious vision into a PROLIFIC business, a series of specific technological objectives will be reached over a 48-month work plan paved with decisive milestones, and ruled by performance indicators enabling the strategic optimization of up to five novel value chains built on still insufficiently tapped biomass.

Objective 1: Mapping the availability and sustainability of the selected feedstocks (legumes, fungi and coffee residues) during and after the project (WP1).

Agro-industrial streams can have an extremely variable composition resulting from *e.g.* seasonal fluctuations, different farming practices and post-harvest processing technologies across EU. An in-depth inventory of the selected residues across EU in terms of amounts, geographical origin, seasonality, and quality variation will be performed. A preliminary assessment of logistic and safety as well as sustainability aspects of the feedstocks will be carried out during the PROLIFIC project and will be refined at project completion. **Performance Indicator:** by month 7 (M1.1), the comprehensive inventory and preliminary assessment report on availability, geographical origin, seasonality, quality, safety and sustainability of feedstocks, is achieved. The collected set of data constitutes key element to improve the further processing of our ingredients and adapt business strategies.

Objective 2: Definition of the specifications and compliance with existing regulations of residue derived compounds (WP1, WP5, WP6 and WP7).

In synergy with regulatory analysis performed both in WP1 and in WP6, the standard requirements of compounds recovered from agro-industrial residues will be specified by end users and proofed by R&D partners for each of the targeted applications. End users will scrutiny advise on the compliance of the foreseen extraction/conversion processes and of the final product prototypes (WP5) with existing regulation (*e.g.* REACH, Novel Foods) as well as with customers' needs as the results of a wise Quality Function Deployment (QFD) (WP7). **Performance Indicator**: by month 7 (M1.1), the target properties as well as the technical and regulatory specifications of proteins and other bio-compounds derived from residues are defined. The information is used to guide and adjust the extraction/conversion and product prototyping processes according to the targeted market applications.

Objective 3: Setting up of flexible and fully scalable biorefinery extraction protocols for the isolation of proteins and bioactive compounds from plant residues (WP2, WP4 and WP5).

Flexible biorefinery extraction approaches will be established and optimised in order to extract proteins and bioactive compounds from selected agro-industrial residues. For the optimization of extraction protocols only industry-driven and fully scalable technologies (WP2, WP3 and WP5) will be considered. These approaches will on purpose be designed for easy application of these new value chains into existing industrial installations. By design, the extraction protocols will be applicable to residues of legumes, fungi and coffee as well as to other streams than the one considered within the PROLIFIC project. **Performance Indicators**: by month 38 (M5.1), protocols are validated at pilot scale and fractions extracted from the 3 streams are fully characterised.

Objective 4: Conversion of extracted fractions into valuable ingredients to specific final applications tailored to the needs of industrial end-users (WP3 and WP5).

The extracts from biomass streams of WP2 will be subjected to different treatments (such as purification, concentration, hydrolysis, and stabilization) and converted into ingredients (*e.g.* peptides) for food, cosmetic and feed formulations (WP5) and components for polymer and composite formulations. Novel food and cosmetic packaging products (WP5) will be based on commercial biopolymers formulations containing bioactive ingredients (WP3) improving the functionality of the material (*e.g.* better preservation). **Performance indicators:** by month 24 ingredients are ready to use for formulation of prototypes; by month 36 (M3.3) at least 4 polymeric formulation protocols are available for further production of packaging prototypes in WP5.

Objective 5: Selection, validation and demonstration of the use of extracted/converted fractions as ingredients in the food sector (WP4 and WP5).

The extracted/converted fractions (WP2 and WP3) will be selected according to industrial end users specifications;

following criteria of favourable safety and techno-economic analyses, positive LCA, and/or an improved functionality compared to benchmark products currently on the market (WP4). Selected food ingredients (*e.g.* proteins/peptides, fibres, β-glucans, polyphenols, carotenoids, phytosterols, caffeine) will be included in the formulation of cereal-based, vegetarian/vegan and meat food products (WP5). The developers and industrial end user will perform sensory consumer and/or panel tests for all food prototypes. **Performance indicators:** by month 32 (M4.1) at least 8 fractions are selected for food applications of which 6 fractions selected for cereal-based, vegetarian/vegan products and 2 for meat products; by month 46 (M5.2) at least 8 food product prototypes are produced, validated and subjected to sensory evaluation and/or consumer acceptability panels in order to evaluate their marketability and their potential as bio-based substitutes to synthetic preservatives in meat products.

Objective 6: Selection, validation and demonstration of the use of extracted/converted fractions as ingredients in cosmetic sector (WP4 and WP5).

According to end users specifications, the extracted/converted fractions (WP2 and WP3) will be selected for cosmetic application following criteria of favourable safety and techno-economic analyses, positive LCA, and/or an improved functionality compared to benchmark products currently on the market (WP4). Selected ingredients (*e.g.* peptides, chitosan, β-glucans, polyphenols, carotenoids, phytosterols) will be validated in cosmetic product prototypes (creams and toothpaste) (WP5) and distributed as give-aways to consumers as a reward for filling of product acceptance questioner. **Performance indicators:** by month 34 (M4.3) at least 2 fractions are selected for cosmetic formulations; by month 46 (M5.2) at least 2 cosmetic product prototypes are produced and validated.

Objective 7: Selection, validation and demonstration of the use of extracted/converted fractions as ingredients/additives in packaging sector (WP4 and WP5).

According to the end users indications, the polymer formulations containing bio-active molecules (polyphenols, phytosterols, carotenoids and caffeine), chitosan, and final residues (WP3), will be characterized in WP4. The best formulations, which satisfy to the requested properties in terms of thermo-mechanical, gas and water barrier, anti-fungal and anti-mites, antioxidant, additive migration, will be used in WP5, for the production of following prototypes: a coating for dry-cured meat, a flexible film packaging for raw and fermented meat products, a rigid jar for cosmetic packaging and coffee capsules. **Performance indicators:** by month 40 (M4.3) at least 8 fractions are selected for packaging; by month 46 (M5.2) at least 4 packaging prototypes are produced and validated.

Objective 8: Selection, validation and demonstration of the use of extracted/converted fractions as ingredients in the animal feed sector (WP4 and WP5).

According to end users indications, the extracted/converted fractions (WP2 and WP3) will be selected for feed applications following criteria of favourable nutritional quality and techno-economic analyses, positive LCA, and/or an improved functionality compared to commercially available benchmark products (WP4). Selected ingredients among proteins/peptides, fibres, polyphenols, carotenoids, phytosterols, will be used in feed product prototypes (pigs and poultry) and validated by *in vivo* studies (WP5). **Performance indicators:** by month 34 (M4.4) at least 2 fractions are selected for feed formulations (1 with improved functional properties and 1 with improved nutritional values); by month 46 (M5.2) at least 2 feed product prototypes are retained and validated.

Objective 9: Demonstration of the safety & regulatory compliance, as well as of environmental & financial sustainability of the developed processes and products (WP6).

In order to ensure the compliance and sustainability of the foreseen products, we will measure the financial and environmental benefits, costs and burdens associated with the new PROLIFIC bio-based products from a life cycle perspective (WP6). Life-Cycle Assessment (LCA) for environmental assessment and Cost-Effectiveness Analysis (CEA) for financial aspects will be applied at an early stage of technology and product development. Furthermore, the compliance of the ingredients and final prototypes with the current relevant regulation, *i.e.* food, cosmetic, packaging and feed will be thoroughly controlled. Safety, quality and purity of the new products will satisfy to EU legislation and will be proven to meet end-market requirements in order to facilitate future market penetration and successful commercialization (Task 6.2). **Performance indicators:** by month 18, preliminary qualitative assessment has been done for a selection of most promising products (M6.1); by month 46, results on product regulations related to final prototyped products are available. In addition, LCA and CEA have been refined for all produced prototypes, demonstrating the sustainability of products and processes (M6.2).

Objective 10: Successful dissemination of project results and preparation for efficient exploitation (WP7).

In order to enhance awareness of the general public as well as of the specialised audience, the overarching objective of the PROLIFIC project is to organise successful dissemination and communication activities via traditional means (website, printed materials, press releases, *etc.*), scientific publications (peer-reviewed journals, conferences, *etc.*), specialised industrial workshops as well as via social media presence. Another important objective is to create an exploitation roadmap consolidating commercialisation plans and business opportunities, which will pave the way

towards successful market entry of the products. Interactive sensory tests and commercial actions with end consumers are integral part of this exploitation roadmap. **Performance Indicators:** by month 48 (M7.2) key performance indicators (*e.g.* organised workshops, participation to exhibitions and fairs, in-store boosts, newsletters, website, publications) listed in Section 2.2 (Table 7), will be achieved.

1.2 Relation to the Work Programme

The PROLIFIC project will contribute to reach following strategic objectives highlighted in the BBI Strategic Innovation & Research Agenda 2017[20]:

1. Foster supply of sustainable biomass feedstock to feed both existing and new value chains

Three different still poorly exploited biomass feedstocks (agro-industrial residues of legume, fungi and coffee processing chains) will be used in PROLIFIC in order to recover proteins, peptides and amino acids, as well as fibres, chitosan, β-glucans and other bioactive molecules (polyphenols, carotenoids, phytosterols, caffeine). The ingredients produced will be used to fill both 5 new (1.hyper-proteic allergen-free vegan food, 2. bioactive and biodegradable food packaging, 3. bioactive and biodegradable cosmetic packaging, 4. bioactive coffee capsules, 5. feed having novel functional and anti-microbial properties) and 4 existing value chains (1. hyper-proteic cereal-based foods, 2. anti-fungal coatings for dry meat, 3. whitening and sunscreen cosmetic products, 4. high nutritional value feeds). Another new development from biorefinery point of view is the targeting of industrially underexploited sources of biomass to mainly recover fibres and polyphenols (from coffee) and proteins and chitosan (from fungi), which are usually produced from fruit and animal sources.

2. Optimise efficient processing for integrated biorefineries through research, development and innovation (R&D&I)

The end users-driven R&D&I activities and product prototyping in industrial/pilot infrastructure will ensure efficient processing of the selected feedstock. The extraction and modification steps optimised and implemented in the PROLIFIC project have been subject to a pre-selection in terms of further realistic applicability at industrial scale, environmental footprint and cost efficiency. The selection of robust and reliable extraction and transformation techniques will facilitate their integration in cascaded biorefinery processes. Their application "in industrial environment situation" will be evaluated in terms of extraction and transformation yields and on the basis of CEA- and LCA results, and whenever necessary be corrected. The R&D&I activities shall also set the ground for the application of these techniques to mixtures of the targeted feedstocks and other sources of biomass.

3. Obtain by means of a flexible integrated approach, a wide array of bio-based products:

PROLIFIC will apply a flexible integrated cascading approach to extract and/or convert bioactive molecules from the three selected biomass processing residues. The selected approach will be scalable in all its steps and potentially applicable to other feedstock different from those used in the project. Unique selling points of the PROLIFIC project are the controllable proteolytic treatment of protein fractions, which will enable the production of a large number of ingredients, and multiple bio-based formulations of polymers with improved functionalities (*e.g.* anti-fungal) that will find application in a wide array of prototyped products and materials.

4. Develop innovative bio-based products for identified market applications

From its conception phase the PROLIFIC project was built around industrial end-user partners that have precise view of market needs in terms of non-meat, meat, cosmetic and feed products as well as packaging materials. In PROLIFIC these industrial end users will drive the development of different types of new food, cosmetic, packaging and feed bio-based product prototypes containing ingredients/additives obtained from selected feedstock residues. This market demand oriented approach is actually the straightest way to pull innovation ground for bio-based products and satisfy customer needs.

The PROLIFIC project addresses the topic "**BBI 2017.R4 – PROTEINS AND OTHER BIOACTIVE INGREDIENTS FROM SIDE STREAMS AND RESIDUES"** as can be seen from the detailed relation of the proposal to the specific challenge and scope addressed by the topic are justified in Table 1.

[20] http://biconsortium.eu/about/our-vision-strategy/sira

Table 1. How PROLIFIC relates to the scope and challenges of the BBI 2017 R4 topic.

Challenges of BBI 2017 R4	How PROLIFIC addresses the challenges
Develop sustainable technologies to recover proteins and bioactive ingredients from feedstock	Green traditional (liquid aqueous extractions) and innovative (ultrasound assisted extraction, microwave assisted extraction, enzyme assisted extraction) technologies will be tested on the raw materials in order to extract proteins (WP2). The biomass residues after protein extraction will also be exhaustively exploited with traditional (alkaline hot water, maceration) and innovative (accelerated solvent extraction, subcritical water extraction, supercritical CO_2 extraction) extraction techniques in order to yield fibres (cellulose/hemicellulose), chitosan, β-glucans and other bioactive compounds (polyphenols, carotenoids, phytosterols and caffeine) (WP2). Final residues will be tested as reinforcement for polymeric composites.
Convert residual biomass and industry side streams into ingredients for food, feed and other high-value markets	PROLIFIC will extract (WP2) and convert (WP3) value added fractions (proteins, fibres, chitosan, β-glucans, polyphenols, carotenoids, phytosterols, caffeine) from legume, fungi and coffee agro-industrial side streams to use them as ingredients/additives in the production of prototypes in the food (cereal-based, vegetarian/vegan, meat), cosmetics (creams and toothpaste), packaging (food coatings, flexible meat packaging, cosmetic jars, coffee capsules) and feed (for pigs and poultry) sectors (WP5).
Achieve separation/extraction and purification (where applicable) of the proteins and bioactive ingredients to meet the required market specifications and regulations.	The project will validate technologies for extracting and converting the components of the agri-food residues for their utilisation in the food, cosmetics, packaging and feed markets. PROLIFIC will focus on improving robust and reliable eco-friendly processes, including the following stages: extraction, fractionation and if necessary purification, ensuring the optimisation of each process and of the overall biorefinery system. Resource/energy expensive purification will be kept to the minimal requirement and preference be given on techniques and applications that enable the use of eco-friendly fractionation techniques. The foreseen value chains are developed "with and for" industrial end-users from the conceptual phase of the project until the production of prototypes. Their knowledge of the market and sector specificities is the best warranty to ensure that constraints related to market specifications and regulations will strictly be considered at each stage of the innovation cycle.
Validate (either at lab scale, or at pilot scale in an industrially relevant environment) a sustainable process to separate and/or convert proteins and bioactive compounds from residual biomass streams originating from biomass production and conversion.	All the processes for extraction of proteins, fibres and other bioactive compounds (chitosan, β-glucans, polyphenols, carotenoids, phytosterols, caffeine) will be tested and validated at lab scale (WP2). When needed, the obtained fractions will be further modified and tailored to targeted formulations and final product prototypes (WP3). All extracts will be tested for safety and bioactivity characteristics (WP4) aiming at applications specific to the food, cosmetic, packaging and feed sectors. The selected fractions and processes (in terms of function, efficiency, sustainability and final ingredient benchmark) will be applied at pilot scale and the resulting fractions and molecules will be used as ingredients/additives to develop and validate final product prototypes (WP5). The piloting and prototyping activities will be carried out in industrially relevant environment in the facilities of the SME and large companies as well as in companies and centres specialized in the biomass valorisation. Processes will be optimised by taking LCA and CEA analysis into account in order to ensure sustainability (WP6).
After separation /extraction and/or conversion processes, ensure that the products' quality and properties are conserved	The starting feedstocks and the fractions and compounds during the processing of biomass will be carefully checked at each level of the cascades of processes. All the fractions obtained after extraction (WP2) and/or conversion processes (WP3), will be analysed by means of a large set of methods dedicated to the assessment of physico-chemical, biological activity and safety characteristics as well as for stability over storage (WP4). Furthermore, the efficiency in terms of increased nutritional value will be scrutiny investigated using a wide array of *in vitro* methodologies as well as *in vivo* testing for feed applications (WP5).
Achieve the appropriate purity of the target molecules and	The purity of the target molecules will be assured by detailed molecular analyses. For protein fractions the total amount measured in bulk way (through nitrogen measurement), the type and amount of released peptides and of amino acids, both

Challenges of BBI 2017 R4	How PROLIFIC addresses the challenges
performance in specific application fields to at least match the benchmark technologies.	bound and free, as well as the protein integrity and identity, will be checked in depth. This will allow to assess the full potentiality of the protein fraction in the foreseen uses, and also to assess eventual obnoxious reactions (*i.e.* allergenicity). The same approach will be followed for the other bioactive compounds extracted. Finally, the presence of possible contaminants of anthropogenic (heavy metals, agrochemicals) or natural (mycotoxins, pathogens) origin will also be assessed in the fractions obtained in order to outline a full risk profile of the new ingredients and prototyped products.
The industry should actively participate to prove the potential for integrating the developed concepts into current industrial landscapes or existing plants so that deployment of the concepts can be accelerated and scaled up to an industrial level.	The PROLIFIC project is exclusively an end user driven action. In fact partners, mainly SMEs and large companies, will take care of the production and/or validation of the final prototype products (COSM, NS, SB, SSICA, FTEN, ILLY). They will instruct R&D partners on the technical specificity requirements concerning the extracted ingredients. The end-users will participate since the very first beginning of the project concerning the definition of final ingredients technical and regulatory standards (WP1), supervising the extraction and conversion used technologies (WP2, WP3), checking the correct assessment of extracted ingredients in terms of quality and safety as well as biological activities (WP4) and using themselves the ingredients in the final products prototyping (WP5). Producers and End users also take largely part to the LCA and CEA analysis (WP6) as well as to the definition of exploitation criteria of the final ingredients and products (WP7). All crucial steps along the created value chains will take place either in the industrial end user facilities or in the technical halls of biorefinery specialists (*e.g.* BBEPP), which do also operate processes applied to other biomass streams at large scale. Taken together with final consumer needs all these measures will pull the developments made within PROLIFIC up to commercially mature prototypes.
Perform a preliminary assessment of the availability of the sustainably produced feedstock used in the project, considering the possible upscaling of the developed processes.	In WP1 we will perform a preliminary assessment of the availability and sustainability of the production of the three selected feedstocks (legumes, fungi and coffee processing residues) during and after the PROLIFIC project. Logistic aspects connected with the project activities, will also be carefully considered. The early involvement of the industrial end users of the fractions and compounds produced will lead to the adequately dimensioning of the processes (WP2 and WP3) as well as streaming the sufficient amount of feedstocks (WP5) in order to adapt the offer to the anticipated market demand and its development potential (WP7).
Specifically demonstrate the benefits of the new developed processes versus the state-of-the-art and existing technologies. Provide evidence of new processing solutions and new products obtained.	The final consumer needs and expectations and the specific requirements of industrial end users manufacturing the prototypes, will set the frame for PROLIFIC R&D activities. In addition, techno-economic progresses made in terms of extraction, modification and conditioning/preserving technologies as well as prototyping, will be benchmarked vs. existing solutions. Emphasis will be put on the comparison of concrete Key Performance Indicators, such as energy- and cost-efficiency, qualitative and quantitative throughput of the products, all aiming at a zero waste concept. A ranking of the new solutions and a final selection will be performed for a later application to pilot scale (WP5). Final consumer-proofed food, feed, cosmetic and packaging prototypes will materialise the progresses made within the PROLIFIC project in terms of new product development.
TRL at the end of the project should be 4-5	All the applied technologies (extraction/conversion and product prototyping) both at laboratory and pilot scale, will be developed at TRL5 or beyond (*e.g.* cereal-based and vegan/vegetarian food products up to TRL 7-8, cosmetics up to TRL 6-7). See Table 2 for detailed analyses of foreseen final TRLs.
An environmental assessment using Life Cycle Assessment (LCA) methodologies, and a cost analysis should be included	The PROLIFIC consortium acknowledged the need to have R&D&I backed up by LCA and CEA to achieve sustainable and price effective value chains built on insufficiently valorised biomass. Therefore, WP6 is dedicated to LCA and CEA starting at an early stage of the project. In depth LCA and CEA of the proposed processes and products will be initiated and constantly refined in order to allow further optimisation of the processes. Data will be collected based on tailored questionnaires from all technological (WP1-4) and demonstration (WP5) and

Challenges of BBI 2017 R4	How PROLIFIC addresses the challenges
	delivered to WP6. The results of these analyses will be discussed with the process developers and owners in order to fully exploit/correct the up-scaling potential of proven/proposed processes.
A viability performance check of the developed process(es) based on available standards, certification, accepted and validated approaches, should be included.	During the final validations of the processes and technological procedures, PROLIFIC will deal with standard GMP (Good Manufacturing Practices) assessments, based on available current standards. These assessments will be based on validation and technological transfer protocols performed for all selected processes and technologies. Food, cosmetic and feed prototype products will follow the strict requirements of GMP-based production, including those standards related to food and cosmetic packaging (*e.g.* HACCP, IFS, BRC, ISO22716).
Results may contribute to establishing new standards for acceptability of products	Food products will be subjected to sensory consumer and/or panel tests (SB, IGV, SSICA) to assess the consumer acceptance of the final products (WP5). A preliminary formulation of cream cosmetic products will be produced and sensorial parameters will be evaluated by a panel test (WP4). During animal trials for feed products an indirect effect on feed sensory characteristics will be observed (WP5). The conclusions of these tests together with QFD results will constitute a solid basis to draft new standards for the acceptability of products (*e.g.* quality) and their acceptance by the final consumers.
Include an analysis of the regulatory requirements and product specifications (like taste,…) that may limit the market approval and acceptance of the targeted ingredients or additives	The final individual ingredients/additives, blends and product prototypes, obtained from the feedstocks, and all related production processes will be analysed both for regulatory requirements, at EU and local levels (WP1 and WP6), as well as for sectorial technical specifications of ingredients and final product prototypes (WP1, WP4, WP5). This will ensure the accomplishment of the EU regulatory framework in each sector and maximize the acceptance by the different markets and targeted consumer groups.
Include specification of the health and safety aspects of the resulting products.	A complete health and safety evaluation of the final extracts produced in the PROLIFIC project will be performed (WP4). An in-depth chemical and biological characterization will allow selecting the best extracts for each application (food, cosmetics, polymers/composites and feed). The biological screening will consist in the safety assessment (toxicity test, mycotoxins, *etc.*) and of several *in vitro* biological activity tests specifically targeted toward the different foreseen final applications (anti-fungal, antioxidant, anti-aging, whitening, *etc.*).

1.3 Concept and methodology

Concept underpinning the project

PROLIFIC is a 4-year Research Innovation Action gathering 17 partners from 7 EU countries and 1 Associated Country. The consortium aims to develop, validate and demonstrate both at laboratory and pilot scale the technical, economic and environmental feasibility of a platform technology for the isolation, conversion and valorisation of value-added molecules from legume, fungi and coffee agro-industrial processing side streams and residues. The extracted molecules: proteins, peptides, amino acids, fibres, chitosan, β-glucans, polyphenols, carotenoids, phytosterols and caffeine as well as residues after sequential extraction will be used as ingredients, as such or after conversion, for product prototyping in food (cereal-based, vegetarian/vegan, meat), cosmetics, bio-based active packaging and feed fields.

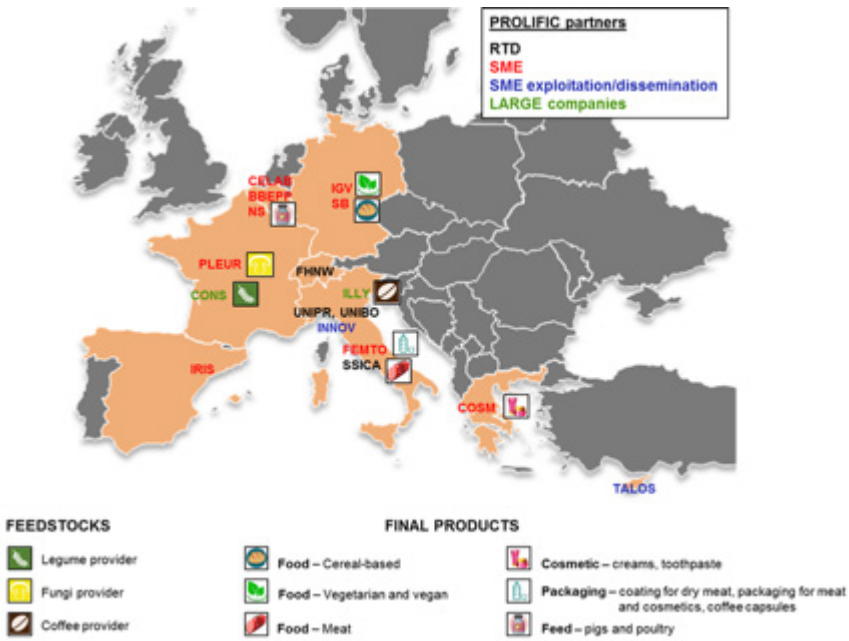

Figure 2. PROLIFIC Consortium, selected agro-industrial residues and final product prototypes

All the extraction, conversion, upscaling, validation and product prototyping processes will be driven by industrial end users, which will use compounds and fractions to manufacture prototype products, which will fully comply with existing regulations (country based, EU and world). Our industrial end users will play a key role in ensuring a direct exploitation of the results and products according to market specifications and benchmarks with similar technologies and products. The developed processes, practices and final products will also be evaluated from a technical, environmental, financial and customer acceptance point of views, thereby contributing to the creation of sustainable value chains in agro-industrial processing sector and pursuing solutions towards 'zero-waste' bio-based operations within the concept of a Circular Economy.

The PROLIFIC project will:

- Optimise laboratory and pilot processes for legumes, fungi and coffee streams by means of flexible sequential extraction and conversion protocols with concomitant validation of the technical feasibility, adaptability and scalability of the developed techniques and to prepare the ground for the valorisation of any type of initial feedstocks;
- Demonstrate the use of each resulting bio-compound fraction in one or more products for the food, cosmetic, packaging and feed application sectors.
- Ensure that the generated knowledge and intellectual property are properly protected and project results are exploited and commercialised based on relevant agreements among the consortium partners.

The industrial product developers and industrial end users of the fractions selected the broad number of market applications (Fig. 2). Their ambition to exploit the innovation potential and economical value of proteins, peptides, amino acids, fibres, chitosan, β-glucans, polyphenols, carotenoids, phytosterols and caffeine obtained from different feedstock sources by using a flexible biorefinery based on sequential extraction/conversion processes. In order to turn this ambition into a reality, the PROLIFIC project will include the compounds and fractions recovered in the formulation of different product prototypes (*e.g.* proteins/peptides as ingredients/additives of food, cosmetics, feed) and finally demonstrate their positive impact on the functionality of the various products, *e.g.* sensory property of vegan products, sunscreen or whitening properties of cosmetic creams (Fig. 1).

Further stakeholders will be involved in an advisory board (SAB). These experienced persons will complement the talent of the partners of the PROLIFIC consortium and will contribute to replicate the action of this project maximizing its impact. The SAB will consist of producers and processors of key feedstocks, food, cosmetic, packaging and feed industries, potential end users of the PROLIFIC ingredients and fractions, association of bio-based industries, policy makers, technology transfer and innovation professionals, investors and consumer associations, cooperatives. They will be consulted as needed for both technical and business related insights.

In PROLIFIC a trans-disciplinary consortium is combining all relevant knowledge, technologies and facilities in order to set the basis of new cross-sectors interconnections and new bio-based value chains demonstrating new applications for different agro-industrial residual streams (Fig. 2).

Why the selected agro-industrial residual feedstocks?

EUROSTAT 2015 data report that the total amount of vegetable waste produced in EU in 2015 accounted for over 3.7 MT[21]. Among those losses a large amount of waste is represented by co-products, residues or side streams of the

[21] http://ec.europa.eu/eurostat

agro-industrial processing pipelines. These streams represent a large amount of biomass still underexploited but rich of high value protein and other bioactive compounds.

The feedstocks were selected in agreement with the industrial partners of PROLIFIC which expressed their needs for proteins, peptides, amino acids and fibres, chitosan, β-glucans, polyphenols, carotenoids, phytosterols and caffeine from the view of the food, cosmetic, packaging and feed sectors. Legumes, fungi and coffee have been chosen considering their potential (bio)chemical composition, content in the various targeted compounds and fractions, volumes of biomass and availability of the biomass for R&D&I purpose up to prototyping.

Legumes as a source of proteins, fibres, carotenoids and phytosterols

The world production of beans accounted in 2014 well over 20 MT (26 MT dry beans, 21 MT green beans)[22] of which 0.7 and 0.8 MT respectively produced in EU. The gross production value estimated in 13.7 and 8.8 billion USD respectively for dry and green beans, worldwide. Regarding peas production, 11 MT for dry peas and 17 MT for green peas were produced worldwide in 2014[12] of which respectively 3.4 MT and 1.1 MT were produced in EU. The gross production value estimated in 3.6 and 13.2 billion USD respectively for dry and green beans, worldwide.

Beans and peas industrial processing by-products generally consist of non-conforming peas, skins and plant parts, especially pods still present after the field harvest. Preliminary data from Conserves France (CONS) estimated the amount of processing residues in 12-14% of the raw material processed. The amount of processing losses from peas and beans in all Europe was quantified in 139,000 and 51,000 T. per year[23]. The general composition of peas by-products was also determined in 80.4% water, 5.6% protein, 0.5% fat, 10.8% carbohydrates, 0.70% minerals[24]. Carotenoids and phytosterol contents in peas, beans and chickpea are respectively about 35-135 mg/100g[25] and 1.4-2.4 mg/100g of fresh weight[26].

Fungi as an alternative source of proteins, chitosan and β-glucans

Mushroom production in 2014 reached about 13 billion T worldwide 6.2 billion T was cultivated in Europe[27]. The mushroom market, in terms of value, is projected to reach over 50 billion USD by 2019, at a CAGR of 9.5% from 2014, with about 13.7 billion T produced and being the fastest-growing food sector in Europe. This is mainly due to consumers' significant change in food preferences and eating habits and the search for low-fat food and health and wellness awareness. Regarding the type of mushrooms, 43% is Paris champignon (*Agaricus bisporus*), 29% is Shiitake (*Lentinula edodes*), 12% is Oyster (*Pleurotus ostreatus*), 16% other mushrooms. The average waste ratio of the mushroom industry is ranging between 5 to 20% of fresh weight and the global waste of fresh mushrooms is 1.87 billion T per year[28]. The general nutritional composition is 20-25% of proteins, 2-3% of lipids, 8-12% ash 8-12% of the dry matter. Soluble and insoluble fibres (chitin/chitosan and β-glucans) account for 30-300 g/kg of dry matter[29].

Coffee as source of fibres, polyphenols and caffeine

Green coffee production worldwide accounted for 8.8 MT[12] none of which grown in EU countries. The gross production value of green coffee in 2014 was about 18 billion USD. The most important EU producer is Brazil followed by Vietnam and Colombia. The vast majority of the worldwide green coffee production is imported by EU countries (about 2.7 MT in 2015). Coffee producers usually submit the green coffee to a roasting process during which several by-products are produced of which the most relevant is silver skin (estimated by ILLY partner to be about 1-2% of the processed green coffee). Silver skin is a fine integument that covers and protects the external layer of the seed, grossly called grain, of green coffee. It is partially taken off in field, during desiccation and removal of coffee seeds from the fruits, and partly during roasting. In 2013, it was estimated a coffee production losses represents about 26,000 tonnes per year in EU[13]. The chemical composition of the silver skin by-product was determined and consists of 70-79% of dietary fibre (15% soluble fibre), cellulose and hemicellulose (around 15%), in about 5% of fat and important bio-functional molecules such as caffeine (0.9%) and polyphenolic compounds (0.6%)[30]. Thus the silver skin may certainly be promising also for the nutritional and cosmetic purposes.

In the PROLIFIC project we will develop cascade of processes targeting major compounds present in each selected feedstock. In addition, in full agreement with the principle of integrated biorefinery and zero-waste philosophy, residual fractions of the three feedstocks, after sequential extraction, will be valorised in the preparation of composite materials for packaging applications (Fig.1).

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[30] T. Gallina Toschi, et al. (2014). Journal of Agricultural and Food Chemistry, 62,10836–10844

Positioning of the project

The activities performed in the PROLIFIC project spans from the technology development/optimisation at laboratory scale (TRL 2-3) to scale up and demonstration of final processes and product prototypes (TRL 5-8). Most technologies are already validated at laboratory and/or pilot scale at RTDs and SMEs facilities and in industrially relevant environments. They are mostly based on prior research of the partners (Table 3). In general, the technologies start at a TRL 3-4, making it realistic to reach up to TRL 5 (and beyond). Table 2 provides a summary of the TRL advances that will be made during PROLIFIC lifespan in terms of targeted final products.

Table 2. Summary of TRL advancements that will be achieved during the PROLIFIC project progression for the ambitioned products

Feedstocks	Isolated fractions	Application category	Application example	Initial TRL	Final TRL
1) Legumes (seeds of peas, beans, chickpea) 2) Fungi (mycelia, processing cuttings) 3) Coffee (silver skin, not compliant roasted seeds)	Proteins, Peptides, Amino Acids	Food (cereal-based, vegetarian/vegan)	– Bread enriched with peptides, amino acids for sportspeople and elderly – Proteins-enriched breakfast cereals and pasta products for children and sportspeople – Protein-enriched vegetarian/ vegan meat surrogates	5 4 3	8 7 7
		Cosmetics	– Acylated peptide fractions as emulsifiers	2	5
		Feed	– Protein-enriched fractions as additives in pig and poultry feed	3	6
	Cellulose/ hemicellulose, chitosan, β-glucans	Food (cereal-based, vegetarian/vegan) Food (meat)	– Chitosan and β-glucans enriched pasta and vegan/vegetarian meat surrogates – β-glucans as additives in cooked meat	3 3	6 6
		Cosmetics	– Chitosan as anti-microbial in toothpastes – Chitosan and β-glucans with water resistant properties in sunscreen creams	4 4	7 7
		Packaging	– Chitosan for dry-cured meat coating (anti-fungal and anti-mites properties) – Chitosan as additive for flexible packaging for meat products – Fibrous extraction residues as reinforcement in composites for cosmetic packaging.	3 3 3	5 5 6
		Feed	– Fibres used as texturing agent or preventing constipation	3	5
	Polyphenols Carotenoids Phytosterols Caffeine	Food (cereal-based) Food (meat)	– Breakfast snacks with polyphenols and/or caffeine – Polyphenols, carotenoids as additives in minced meat products	3 3	6 5
		Cosmetics	– Polyphenols, carotenoids or phytosterols in creams as anti-ageing or skin whitening actives	3	6
		Packaging	– Polyphenols, carotenoids, phytosterols and caffeine as	3	5

Feedstocks	Isolated fractions	Application category	Application example	Initial TRL	Final TRL
			bioactive additives in packaging for cosmetic and in coffee capsules		
		Feed	– Carotenoids as additives in pigs and poultry feed – Polyphenols as functional feed ingredients (anti-microbial, antioxidant and anti-inflammatory activities)	3 3	5 5

National or international research and innovation activities linked to the project

The present project capitalizes on prior state-of-the-art, knowledge and networks developed in on-going (BBI and H2020) or very recently terminated EU projects (FP7) in all the range of disciplines relevant to PROLIFIC multi-disciplinary approach and structure. A selection of relevant projects in which of project partners are/were involved (further projects can be found in the respective partner profiles) is reported in Table 3.

Table 3. National or international research and innovation activities linked to PROLIFIC. WPs which will benefit of the knowledge of listed projects are indicated. WP7 will benefit from inputs from all these projects.

Project name and description	Partners involved	Activities performed and relevant for PROLIFIC	WPs
INMARE (H2020, 2015-2019) Innovative screening and expression platforms to discover and use the functional protein diversity from the sea.	FHNW	Development/optimization of immobilization of oxidases and esterases onto nanoparticles and protection of the biocatalysts by means of controlled layer of organo-silanes	3
MADFORWATER (H2020, 2016-2020) Development and application of integrated technological and management solutions for wastewater treatment and efficient reuse in agriculture tailored to the needs of Mediterranean African Countries.	FHNW, UNIBO	Lifecycle Assessment (LCA) and Cost Benefit Analysis (CBA) for new innovative water treatment and irrigation technologies still under development and immobilization of enzymes onto nanomaterials and other support material.	6
AgriMax (BBI, 2016-2020) Agri & Food waste valorisation co-ops based on flexible multi-feedstocks biorefinery processing technologies for new high added value applications	IRIS (coordinator), UNIBO, SSICA, FTEN	New applications for the agricultural value chains by applying processing for valorising crops and food wastes.	1-6
BIOSEA (BBI, 2017-2020) Innovative cost-effective technology for maximizing aquatic biomass-based molecules for food, feed and cosmetic applications	IGV	Extraction, concentration and purification of target bioactives and new product development.	1-5
BARBARA (BBI-JTI, 2017-2020). Biopolymers with advanced functionalities for building and automotive parts processed through additive manufacturing	CELAB	Valorisation of side-stream fractions and residues from agro-food production into novel polysaccharides and functional additives.	3- 5
AFTERLIFE (BBI-JTI, 2017-2021). Advanced Filtration TEchnologies for the Recovery and	CELAB, BBEPP	Flexible, cost- and resource-efficient process framed in the zero-waste and circular economy	2-4

Project name and description	Partners involved	Activities performed and relevant for PROLIFIC	WPs
Later conversIon of relevant Fractions from wastEwater.		approach for the recovery and valorisation of the relevant fractions from wastewater.	
PULP2VALUE (BBI-JTI, 2015-2019). Processing underutilised low value sugar beet pulp into value added products	BBEPP	Development of multiple extraction techniques to isolate more valuable products (microcellulose fibres, arabinose and galacturonic acid) from sugar beet pulp.	2-5
Nano3Bio (H2020, 2014-2017). NanoBioEngineering of BioInspired BioPolymers	BBEPP	Development of recovery methods for fermentatively produced chito-oligomers, and a view on their application potential.	2-5
REHAP (H2020, 2017-2020). Systemic approach to Reduce Energy demand and CO_2 emissions of processes that transform agroforestry waste into High Added value Products.	BBEPP	Development of extraction technologies to recover tannins and fibres from forestry waste	2-5
NOAW (H2020, 2016-2020). No Agro-Waste: Innovative approaches to turn agricultural waste into ecological and economic assets.	UNIBO	Development of a circular economy approach applicable to agricultural wastes, by converting them in bio-energy, bio-fertilizers, bio-packaging and bio-molecules	2-5
InDIRECT (H2020, 2016-2019) Direct and Indirect biorefinery technologies for conversion of organic side-streams into multiple marketable products	UNIPR, NS	Development of biorefinery processes for the conversion through insects of diverse organic side streams into proteins, chitin and lipids. Testing and validation of functional feed ingredients for animal feed applications	2-5
TRUE (H2020, 2017-2020) Transition paths to sustainable legume based systems in Europe.	IGV	New breeding strategies of legumes resulting in products with higher protein content and tailored amino acid ratios	1-5
CITRUSPACK (LIFE, 2017-2020) Valorisation of citrus fruit waste into a series of new value chains.	COSM	Use of citrus remainings and fibre for cosmetic products and packaging.	2-5
BIORICE (FP7-SME, 2013-2015) Production of bioactive ingredients from protein by-products contained in the processing water of the rice starch production stream.	UNIBO (coordinator)	Enzymatic and microbial treatment of protein by-products, isolation of peptide fractions, characterisation of their bioactivity and safety and integration into trial nutraceutical and cosmetic products.	2-6
PROSPARE (FP7, 2007-2011). Progress in Saving proteins and Recovery of Energy	UNIPR (coordinator)	Reuse of poultry leftovers through hydrothermal degradation and enzyme processing to obtain new proteic material to be used for food and feed	2, 3, 5
NOSHAN (FP7, 2012-2016) Production of Safe and Functional Feed from Food Waste	UNIPR, IGV, NS	Characterization of vegetal and dairy food waste streams and development of processes for their valorisation as bulk feed and feed ingredients	1-5
APROPOS (FP7, 2012-2014) Added value from high protein & high oil industrial co-streams	IGV	Use and conversion of co-stream fractions from oily crops residues and fish filleting residues into new food, feed and textile products	1-5
ProVegFlake (FP7-SME, 2016) High grade PROteins innovatively processed into VEGetal FLAKEs from plant sources with enhanced nutritional benefits	IGV (coordinator)	Development and market launch of tailored protein products for elderly, sportspeople	4-5
PHYTOME (FP7-SME, 2012-2015) Phytochemicals to reduce nitrite in meat products	SSICA	Development of new meat processing technologies and innovative products enriched	4-5

Project name and description	Partners involved	Activities performed and relevant for PROLIFIC	WPs
		with selected plant bioactive compounds to reduce nitrite levels and improve health.	
PROCURED (FP7-SME, 2014-2015) Optimization of the salting process for the production of healthier and higher quality dry-cured meat products with reduced and more standardized salt content	SSICA	Development of a new salting control platform to enable a non-destructive estimation of lean/fat content in raw meat to select processing technologies suitable for the absorption of polyphenols.	4-5
LEGUVAL (FP7-SME, 2013-2016) Valorisation of legumes co-products and by-products for package application and energy production from biomass	IRIS (coordinator), SSICA	Development of a process for the extraction of proteins and fibres from different kinds of legumes by-products at lab and pilot scale	2-5
BIOCOPACPLUS (LIFE+, 2014-2016) Sustainable bio-based coating from tomato processing by-products for food metal packaging	SSICA (coordinator)	Development of a method for the extraction of cutin from tomatoes by-products at semi-industrial scale; use of cutin for applications in food packaging sector	3, 5
SUBWEX (ERA-NETs Cornet, 2013-2015). Subcritical water as a green solvent for extraction of plants.	CELAB	Developing an integrated process chain to produce high-added-value additives for the food, nutraceuticals and cosmetics sectors and at feedstock for the paper industry.	2-5
MICROFLUSA (H2020, FET-OPEN, 2015-2019). Fabricating colloidal materials with microfluidics	TALOS	Dissemination and exploitation manager	7
CY-Biobank (H2020, Teaming, 2015-2016). Biobanking and the Cyprus Human Genome Project	TALOS	Dissemination and exploitation manager	7

Other relevant ongoing projects in which partners are not involved and with which contacts will be initiated for the organization of a joint Stakeholder event (WP7) are: FUNGUSCHAIN (BBI-VC3 agro-based value chain) which aims to produce bio-based functional additives and biopolymers from mushrooms offcuts; GREENPROTEIN (BBI-VC3) which will extract functional proteins such as Rubisco from vegetable processing industry remnants to use it as food ingredients; PROMINENT (BBI-VC3) which aims to extract food grade proteins from cereal processing residues; MACROCASCADE (BBI-VC3), MAGNIFICIENT (BBI-JTI) and ABACUS (BBI-JTI) which aim to extract added-value proteins and other bioactive compounds from algal biomasses to be used as ingredients in feed, food and cosmetics; BIOSMART (BBI-JTI) which will develop active and smart bio-based and compostable packages addressing the needs of fresh and pre-treated food applications; SSUCHY (BBI-JTI) which aims to develop innovative processes and constituents for structural and multifunctional bio-based composites starting from lignocellulosic feedstocks. In addition, during Task 7.2 of the work plan, we will build links with numerous other synergistic projects and initiatives from H2020 (*e.g.* SC2 and SC5 calls) as well as national and regional schemes for exchange of information and best practices. In addition, PROLIFIC will engage with the European Innovation Partnership (EIP) "Agricultural Productivity and Sustainability", which aims to provide a working interface for innovation actors, including farmers, advisors, agri-business, civil society, and researchers, working at EU, national and regional level to act as a catalyst for innovation-related actions to foster enhanced productivity and sustainable resource management across the whole value chain. In addition, we will also link to the activities and working groups of EIP-RAW MATERIALS, as well as to the SPIRE PPP. The consortium will also interact pro-actively with European Technology Platforms, *e.g.* SUSCHEM, bio-based economy stakeholders, *e.g.* Europabio and Nova Institute, and associations of specialists of bioprocesses *e.g.* European Federation of Biotechnology and Asian Federation of Biotechnology.

Overall methodology

The scarcity of conventional sources (plants and animals) and the negative environmental impact of their production/cultivation require R&D&I intensive efforts for alternatives that are environmentally and economically sustainable. The PROLIFIC project focuses on the food, feed, non-food and cosmetic industrial sectors and targets the main compounds used in these sectors. The PROLIFIC methodology is exclusively industry-driven and is centred on **two core components:**

1) Industry-driven feedstocks extraction, conversion and final fraction characterization

These activities will be carried out within the four research and innovation work packages, in which valorisation of initial feedstocks towards final applications will serve as a red thread and will be implemented by means of close interaction of industrial end-users with R&D partners. In WP1 we will characterize the biochemical properties and safety relevant issues the feedstocks. In parallel, we will assess their overall availability and the sustainability of the foreseen processes/activities. WP1, 2, 3 and 4 are intimately interconnected in order to isolate, according to the existing regulations, and convert ingredients/fractions following the specifications and requirements of end users for future applications (food, cosmetic, packaging and feed) as well as market requirements and consumer needs (QFD).

2) Product prototyping and innovation, market analysis and exploitation

Starting from QFD results, PROLIFIC products will be developed to match final consumer expectations under the guidance of industrial end-users of the compounds and fractions recovered in the frame of this project. Some activities will transversally accompany the R&D work and aim at demonstrating the effective integration/application of isolated/transformed fractions as ingredients/additives in food, feed, cosmetics and packaging final products (WP5). LCA and CEA methodologies (WP6) will also be deployed all along the project activities in order to demonstrate the sustainability of PROLIFIC prototyped products. Feasibility assessment will be performed for each product and will serve a basis for the performance of LCA at the early stage of the project. In parallel, a CEA will be performed for PROLIFIC products including estimation of the cost components as well as a financial evaluation for future commercialisation. Another major activity will be the preparation for exploitation and commercialisation of the PROLIFIC products (WP7) with the aim to ensure that partners will be in the position to enter the market with these new products in the post-project phase. Relevant activities, such as IPR protection, knowledge management and the preparation of the Exploitation Agreement among the partners will be performed in WP7. Technical activities such as the prototyping and evaluations (sensory panels for food and *in vivo* trials for feed) (WP5) and marketing actions like the distribution of bio-based give-aways to European citizens against their feedback will constitute first indicators on market penetration and commercial successes.

How sex and/or gender analysis is taken into account in the project's content.

The PROLIFIC consortium takes a comprehensive approach towards gender aspects, which includes gender mainstreaming and positive actions. In accordance with the Articles 2 & 3 of the Treaty of Amsterdam (1997) and other EU policy directives (COM (96) 67 final) and reports (EUR 2002), the consortium is committed to follow the principles of gender mainstreaming. The project is gender-neutral and will give the same job opportunities to everyone, based on their competencies and skills and without any segregation based on sex/gender issues. Gender is relevant to the content of the project because on the one hand the presence of females in the agricultural sector is traditionally limited and on the other hand there is a strong presence of females along the value chain of the bioeconomy industries. To this end, innovation will consider the gender dimension and cater for the needs and motivations of females where possible. This is important given that female professionals play an increasingly prevalent role in decision-making in the food processing, cosmetics, packaging and in general in bio-industries. This will be highly relevant in the consumer perceptions study regarding the PROLIFIC products prototypes. It is noteworthy that there is a strong presence of female researchers, technologists, and industrials in the consortium, a number of which are occupying key positions in the project including in the Scientific Coordinator, Project Steering Committee, the Valorisation Committee and in WP and task leadership roles. A recruitment policy for future researchers and developers for the project is in place and is designed to ensure a reasonable level of balance between the sexes. In order to foster the future of the circular Bioeconomy in Europe we will act for the education and training of the next generation of scientists, technologists and others jobs needed in the EU.

1.4 Ambition

PROLIFIC has the ambition to **demonstrate the technical and economic feasibility, the environmental sustainability** of extraction/conversion processes from selected biomass as well as the suitability of the fractions to be used as ingredients/additives in prototypes of food, cosmetic, packaging and feed products **in order to support the circular economy and increase co-products/residues valorisation practices**. Our final product-oriented ground-breaking objectives will lead to large knowledge contributions to advance the current the state-of-the-art.

1) **State of the art and progresses in the field of co-products/residues processing/conversion industrial technologies**

In current industrial practice, side streams are more of a cost rather than a value. Disposal of food side-streams usually represents a problem that is further aggravated by legal restrictions. Thus, the valorisation of these biowaste by a single process producing high valuable compounds is a good solution. In general, the valorisation practice is to use the side streams to spread on the land as a fertilizer or for biogas production. The latter is common practice is countries like Germany, Belgium or UK. Side streams rich in protein (over 30% content) can be sold directly as cattle feed. Due to the high perishability, this is only possible for a number of days. A solution to solve this problem is to dry the side stream in pellets. However, this practice is "energivorous" and is therefore costly and only applied to streams with high protein content. Only once in a while the extraction of valuable compounds from side streams and use for commercial application reaches demonstration phase and enters later into production. Examples currently in production phase are the use of gluten from cereals (*e.g.* Cargill with the products GluVital and ProwLiz) or whey from cheese production (*e.g.* Frieslandcampina DMV in the brand Textrion PROGEL) as protein sources. An example of projects currently in demo phase is the BBI-JU project Pulp2Value, where microcellulose fibres, arabinose and galacturonic acid are extracted from sugar beet pulp and used for food, personal care and chemical products. Other BBI-JU projects, *e.g.* the projects Prominent and Newfert that are extracting side streams from wheat and rice for food or from biowaste for fertilizer applications respectively, are still in research phase. The extraction of very valuable compounds, for the use as nutraceutical or for the flavour and fragrance industry, are more common practice but still limited to niche markets. Examples are the Belgian SME Ecotreasures that extracts antioxidants from fruit-seeds by using supercritical CO_2, or the extraction of the fragrance molecule valencene from citrus peels.

In PROLIFIC we selected side streams with substantial volume in a single location or where logistics for concentrating the side stream into central biorefineries is relatively simple to set up. These feedstocks are currently under-valorised. The applications are also markets with high volumes. The prototyping of products under guidance of industrial actors and with feedback from final consumers is promising as to future market penetration. PROLIFIC partners are optimistic in achieving another new success story built on the valorisation of insufficiently tapped biomass by optimizing new approaches that are more cost-effective than current extraction technologies.

Protein extraction process consists of (i) mechanical grinding and/or pulverization of the feedstock to make it more accessible for the following treatments, (ii) protein extraction from the matrix, (iii) protein isolation/conversion, (iv) protein formulation. *PROLIFIC will go beyond the state of the art in protein extraction from the matrix with the following technologies.* Regarding protein extraction in legumes, different approaches are currently applied based on the use of aqueous solutions[31]. In the majority of cases, the extraction is carried out either at alkaline or acid pH, often at high temperatures[32]. However alkaline or acid hydrolysis tends to be difficult to control, and yield products with modified amino acids. The use of proteolysis can be useful to increase protein extraction yields. Enzymatic hydrolysis can be performed under mild conditions as opposed to the extreme environments required by chemical treatments. Usually, enzymatic processes prevent side reactions and do not decrease the nutritional value of the protein source. In addition, enzymes present substrate specificity, which permits the production of protein hydrolysates with better-defined chemical and nutritional characteristics. However, enzymatic processing can present certain limits. For instance, total protein hydrolysis by one protease action is still utopic, but a large range of protease variants is currently available, each with its own specificity. Enzyme application is also limited by the relative instability of the biocatalysts, but the decay of enzymatic activity can be circumvented using immobilized and protected forms of the enzymes[33]. The application in magnetic or membrane reactor facilitates the recyclability of the immobilised enzyme.

In PROLIFIC we will take advantage of the knowledge generated in other projects (e.g. Leguval, InDirect, NOSHAN, Table 3) with the aim of obtaining extraction yields of protein extracts (in particular from legumes) up to a 80% degree of purity. We will further develop an enzyme-assisted extraction method in neutral conditions, with the use of phosphate buffers containing high amounts of salts[34]. We will also optimise the use of isotonic salt solutions in

[31] B. Tiwari, N. Singh (2012). Pulse Chemistry and Technology, *RSC Publishing*
[32] M.A. Monsoor, H.K.M. et al. (2002). Int. J. Food Science and Technology, 37, 97-99.
[33] MR Correro et al. (2016) Angew Chem Int Edit 55, 6285–6289.
[34] M. Fredrikson et al. (2001), J. Agric. Food Chem., 49, 1208-1212

combination with enzymes. The technology is currently at TRL 6. With different water-based solvents (isotonic salt solution) and the use of enzymes, the protein content of extracts can be increased without using precipitation agents. In processes making use of proteases, these we be will immobilised at the surface of nanoparticles and protected by a layer of organo-silane. The technology is currently at TRL 3 and we will bring it to TRL5.

Regarding step (iii) protein isolation and step (iv) formulation: filtration methods are mostly used at industrial scale. These methods enable the separation of salts from the proteins. An alternative and more traditional method is based on isoelectric precipitation[35]. Concentration using vacuum evaporation is proven at TRL 9 and will be optimised in PROLIFIC. Ultrasound Assisted Extraction (UAE) is a flexible process intensification approach able to reduce extraction times and solvent use as well as improve protein recovery yields and product quality. UAE is beneficial in case of recalcitrant matrices (*e.g.* coffee residues) to recover not only protein but also bioactive compounds.

In PROLIFIC we will assess the optimal conditions for UAE of high added value compounds using a dynamic system to test ultrasound frequency, wavelength and amplitude as well as solvents properties. The aim of these trials is to develop a system able to process the different raw materials at high efficiency, taking advantage of the flexibility of the UAE, thus resulting in an extraction step with low investment and low operational cost.

Concerning the non-protein fractions of side streams material, a clean technology that is currently under development in Europe is the subcritical water extraction (SWE) or pressurized hot water extraction[36]. This technology applies heat and pressure to water to lower its polarity, causing it to behave like an organic solvent. Compared to conventional extraction process, SWE is not toxic (organic solvent free process) and the operation times are shortened. To our knowledge, only one company (the Mazza Innovation Ltd, Canada) is already using this technique (TRL 9). Pressurized hot water can also be used as reaction media[37]. The applied conditions change the properties of water (higher concentration of hydronium ions), which promote the hydrolysis reaction. This property can be used to hydrolyse polymers in milder conditions than the traditional ones, the polymers therefore become soluble but are less degraded. SWE is a fast and easy technique to implement, only the temperature and operation times need to be optimized. Even though, more studies are required in order to evaluate all aspects of environmental impact assessment. The PROLIFIC project will partly fulfill this need by means of LCA.

PROLIFIC project aims to promote this green process in Europe and to investigate the SWE process for a much wider range of applications than only extraction of small compounds for food, dietary supplements and cosmetic, such as hydrolysis of polymers (cellulose/hemicellulose, chitin) for biobased material.

2) State of the art regarding price of similar compounds and fractions on the market.

Break-through of novel technologies and products is mainly ruled by the market demand and the best price-quality ratio. Considering the demand for our products is given and anticipating significant progresses in our technology development, pricing of compounds and fractions produced by the PROLIFIC consortium was made with our industrial end-users. Our future compounds and fractions have been benchmarked with similar ingredients on the market (Table 4).

Table 4. Current applications and benchmark of targeted molecules and expected values from PROLIFIC resulting compounds

Extracted compounds	Common source feedstock	Current applications	Actual bench-mark, commercial price (€/kg)	PROLIFIC applications	PROLIFIC projected bench-mark (€/kg)
Proteins	Pea, hemp, lupin, wheat, soy Soy meal	Food (cereal-based) Conventional protein source in animal feeds	5-8 0.5-1	Bread, breakfast snacks, pasta, vegan/vegetarian Animal feed	2.5 0.35
Peptides/ amino acids	Wheat peptide +olive oil Soy meal	Cosmetic emulsifier Protein source in animal feeds	20 1-1.5	Food (breads) Cosmetic creams Animal feed	12 12 0.9
Cellulose/ hemicellulose	Wood by products	Functional fibre	1.1-2	Packaging Animal feed	1 0.8

[35] L-P Des Marchais (2011). Procedia Food Science, 1, 1425-1430
[36] M. Castro-Puyanas et al. (2017). Current Opinion in Green and Sustainable Chemistry, 5:31–36
[37] O. Benito-Roman et al. (2016). Int. J. Carbo. Chem., 1-6

Extracted compounds	Common source feedstock	Current applications	Actual bench-mark, commercial price (€/kg)	PROLIFIC applications	PROLIFIC projected bench-mark (€/kg)
Chitosan	Crustaceans	Agriculture, cosmetics, food preservation	50	Pasta, vegan/vegetarian Cosmetic creams and toothpaste Meat food coating and packaging	20-40
β-glucans	Oat Yeast Yeast	Food Cosmetics Immunostimulants in animal feed	35 1.6 1.1-2	Cereal-based/ vegan vegetarian, meat food texturisers Cosmetic cream Animal feed	20 0.7 0.9
Polyphenols	Grape, fruits Grape, fruits Grape	Additives in food Cosmetics Antioxidants in feed	30-80 80 1.4-2	Food Cosmetic creams Packaging Animal feed	25 55 55 0.9
Carotenoids	Microbial synthesis	Antioxidants in food, cosmetics, feed	1-2	Food (cereal-based and meat) Polymer additives Animal feed	1 1 0.7
Phytosterols	Avocado	Cosmetics	40	Cosmetic creams Polymer additives	30 30
Caffeine	Coffee	Food additive Cosmetic enrichment, Beverages	28 33 22	Cereal snacks Cosmetics Polymer additives	20 25 20

PROLIFIC targeted molecules and fractions, obtained by means of eco-sustainable extraction and conversion methodologies will match or be below the benchmark prices of commercially available equivalent compounds and fractions.

3) State of the art regarding foods including ingredients similar to those produced in PROLIFIC

Cereal-based products

The market for functional food and drink has been growing rapidly in recent years and consumers are beginning to accept these products as part of their daily diet. Cereals, bread, pasta and nutritional bars are being fortified with vitamins, minerals and protein in order to be more attractive to consumers seeking a healthier lifestyle. According to data from Euromonitor[38], the fortified and functional food market, including packaged food, was valued at 159 billion USD globally in 2016, making it the largest category within the health and wellness industry (H&W). In particular, high-protein foods started out being popular among fitness enthusiasts due to the role they play in muscle repair, but they are now also widely used for their anti-aging, cardiovascular health and mainly weight loss benefits. Proteins induce satiety therefore foods fortified with proteins mainly have a weight management positioning, valued at 93.5 billion USD globally in 2016. Dairy products are by far the most popular foods fortified by proteins (*e.g.* Greek yogurt by Chobani). In addition, savoury snacks, bread and pasta are increasingly being fortified with protein. Meat-based snacks are also becoming more popular, as they are naturally high in protein. Euromonitor Survey[24] revealed that products that are naturally functional have a great potential among consumers' preferences. Natural healthy food is valued at 101 billion USD globally in 2016, making it the second largest category in health and wellness, behind fortified/functional food, with a predicted 18% increase in value by 2021. This increasing demand for natural products is very much related to the clean label foods trend, which is valued at 130 billion USD globally in 2016 (28 billion USD for all-natural labelled foods alone). Always more consumers like the idea of plant-based foods with

[38] Euromonitor International's Global Consumer Trends Survey, "What is important to consumers purchasing food and beverage products, www.euromonitor.com

intrinsic protein, vitamin or mineral content and no artificial fortifying additives. Vegetable-based savoury snacks and biscuits are also following this trend (*e.g.* Good Thins line by Mondelez International).

The new protein-based extracts developed by PROLIFIC will be used to enrich the food products of IGV and SB, which are currently medium scale food-specialities. Moderate protein contents will be combined with selected bioactive compounds (e.g. polyphenols, carotenoids) suitable for human nutrition. A key feature will be the production of tailor-made batches of specific cereal-based products for special clients-needs demanding new designs in terms of shape, structure, taste, natural colour and odour. The existing facilities of the partners (extraction and extrusion) are suited to gain at pilot scale protein enriched extracts from legumes (chickpea, beans, peas), coffee and fungi residues. For the use in breakfast food and bread products the protein contents of these extracts will be increased. The added-value functional bread technology of SB is positioned at TRL 6. Technological optimizations will be performed during PROLIFIC implementation in order to achieve TRL8, whereas the aim will be to create a new technological procedure which does not alter the functionality nor the nutritional quality of the active fractions used for wholemeal breads.

Vegetarian and vegan products

For the increasing world population from 6.9 billion in 2010 to about 9.5 billion by 2050 new trends in human nutrition are observed utilizing protein-rich plant resources, such as soy, and other leguminous seeds, algae or insect proteins[39] [40] [41]. The market forecast for such protein ingredients has been reported to grow at a compound annual growth rate of 6.0% from 2017 to reach 58.49 USD billion by 2022[42]. The rising market for nutritional food focused on vegetal protein ingredients is ascribable to the major trend among customers to reduce meat consumption for health and sustainability reasons. These so-called "flexitarians", compose for example in Britain 29% and in Germany 37% of the customers[43],[44]. Primarily, flexitarian customer food interests are focused on a clean label based on transparency of labelling (*e.g.* authenticity of ingredients), sustainability of food supply, free of allergens, absence of GMOs or synthetic additives as well as relevance to nutrition and consideration of the health Claims[45]. Data from Innova Market Insights Report have shown a 63% CAGR rise in the number of global food and beverage launches using vegetarian-flexitarian lifestyles[46]. In the EU approximately 5% of the population cultivates a vegetarian lifestyle[47]. In addition, many consumers have aversions to soy or wheat, so that their options to achieve a balanced nutrition are being shortened and orientated on health-promoting products. Proteins could be characterised by their nutritional value, deduced from the essential amino acid-sequences, in which meat proteins are usually nominated with 100%. Most vegetal proteins are classified with bioactive nutritional values between 50 to 90% from which soybean protein has a high value of about 86%. Combining proteins is therefore a necessity in a vegan and vegetarian diet[48].

PROLIFIC protein-enriched extracts will allow producing vegetarian/vegan food products with proteins and essential amino acids to reach the needed nutritional value upgrade[49]. Many products of daily use, especially convenience goods, will be complemented, and consumers will not have to change their priorities of taste nor use supplement products when using PROLIFIC products. PROLIFIC protein-based products will also take advantage from the "free of allergens" label positioning, when compared with direct competitors based on soy-proteins, wheat proteins or egg-proteins. The developed food products will be competitive in terms of value for money providing substantial market advantage for the partners of PROLIFIC. In conclusion, based on selected feedstocks and optimised technologies PROLIFIC products will go beyond current state of the art and will have all of the following main characteristics: exclusive use of vegetal protein, absolutely vegan, allergen-free, gluten-free, GMO-free, high protein content often over 50% (highest is 85%).

Meat products

Meat and meat products supply essential amino acids, Fe, Zn, Se, vitamin B6 and B12 and vitamin D. On the other hand, in 2015, the International Agency for Research on Cancer (IARC) asserted that there would be a statistically significant dose-response relationship between the colon-rectal cancer and the consume of red meat[50]. The major

[39] UN (2012). United Nations Department of Economic and Social Affairs/Population Division and World Population Prospects, 2012
[40] Spiegel M. et al. (2013). Comprehensive Reviews in Food Science and Food Safety 12.6 (2013): 662-678.
[41] Petrusán, J.I. et al. (2016). Current Topics in Peptide & Protein Research, 17, 1-19
[42] MARKETS AND MARKETS (2016): Meat Substitutes Market by Type (Tofu & Tofu Ingredients, Tempeh, Textured Vegetable Protein, Seitan, Quorn), Source (Soy-based, Wheat-based, Mycoprotein), Category (Frozen, Refrigerated), and Region - Global Forecast to 2022.
[43] Derbyshire E.J. (2017). Front. Nutr. 3:55. doi: 10.3389/fnut.2016.00055.
[44] GFK, Consumer Panels, Total Grocery 2016, IMMER SCHÖN FLEXIBEL BLEIBEN: Ein gutes Drittel der Haushalte sieht sich als Flexitarier – der Anteil steigt (aus Gesundheitsgründen) mit dem Alter
[45] Regulation (EC) No 1924/2006, lastly amended by Commission Regulation (EU) No 1047/2012; Commission Regulation (EU) No 432/2012
[46] Innova Market Insights, 2016
[47] European Vegetarian Union EVU e.V.; VEBU Vegetarierbund Deutschland e.V.; www.statista.com
[48] Sharon Palmer, RDN: Protein in Vegetarian and Vegan Diets, RD Resources for Consumers 2014 www.vegetariannutrition.com;
[49] González A.D. et al. (2011). Food Policy, 36, 562–570
[50] International Agency for Research on Cancer (IARC), (2015). http://dx.doi.org/10.1016/S1470-2045(15)00444-1.

promoter of colon carcinogenesis in the red meat is the heme iron, via N-nitrosation or fat peroxidation[51]. These pathways may be suppressed adding ingredients enriched in antioxidant and anti-nitrosant molecules (phytochemicals). Many flavonoids and other bioactive compounds found in plant extracts were shown to protect against meat oxidation, which is an important cause for off flavouring[52]. In the PHYTOME EU project (SSICA partner), meat products enriched in bioactive molecule were manufactured by combining the addition of natural extract mixtures with changes of the processing techniques. The early results showed that the new meat products had more antioxidant capacity and a better oxidative stability than the conventional ones and that their consumption reduced the formation of nitroso compounds in human body. As the excess fat may be a pro-cancer factor in red meat and processed meats[53] as well as cause chronic diseases[54], the fat reduction is one of the challenges to make healthier processed meats. Various health benefits are associated with soluble fibres (β-glucans), *e.g.* serum cholesterol lowering effects[55],[56]. In recent years, many fat replacers have been tested in emulsified-type meat system, beef patties and fermented sausages[57]. The fat replacement with soluble fibres improves the nutritional profile and the tenderness of meat products and increases the yields due to the highly viscous nature and water-binding, foam and emulsion stabilizing capacities of the fibres[58].

Based on previous results and observations, new PROLIFIC extracts enriched in polyphenols and carotenoids and having ascertained functional activities, will be added into minced meat products (e.g. hamburgers, fermented sausages) to improve the antioxidant capacity and oxidative stability, preserving at the same time the sensory profile of the products. Besides, soluble fibres (β-glucans) extracted from selected feedstocks, will be added into a cooked meat sausage to reduce the fat content and improve the nutritional profile and preserving sensory traits and textural properties. In the PROLIFIC project, partners will optimize the manufacture of healthier meat products helping to expand the market and to increase the competitiveness of the meat industry.

4) State of the art regarding cosmetics including ingredients similar to those produced in PROLIFIC

There is a strong trend towards the use of sustainably produced raw materials in cosmetics. There are annual events like the Sustainable Cosmetics Summit[59] to promote their use. Even big companies like L'Oreal are tracing and measuring the sustainability of the ingredients used in their cosmetics[60]. The production of ingredients/additives extracted from residues of renewable feedstocks in PROLIFIC will help to increase sustainability.

Cosmetic marketing is often based on the beneficial properties of a plant and its traditional use. High-protein plants are most commonly used as starting material for producing protein concentrates (*e.g.* wheat, corn, soy, rice and oat), and defatted oils (*e.g.* almond). Among the large variety of vegetable proteins, wheat gluten and soy globulins are by far of the widest used because of their high elasticity when hydrated (wheat) or of their gelling and emulsifying effects (soy). To make proteins suitable to be incorporated into water-based cosmetics, they need to be converted into soluble form and this achieved by hydrolysis. Several skin and hair care formulations include proteins as they create a suitable environment for healthy skin and hair because of their ability to bind water. Proteins within cosmetic formulations are also valued for their strong moisturizing and film forming properties. Many proteins that have been used throughout the years in the cosmetic industry were derived from animal sources. Other plant-derived bioactive compounds are usually included in cosmetic formulations such as olive oil and phytosterols as emulsifiers or chitosan (from crustaceans) as water-resistant agent. Currently vegetal fibres are deployed for cosmetics applications. Polyphenol extracts (*e.g.* from grape and green tea) are attractive ingredients for cosmetics and pharmacy due to their beneficial biological properties. Also the number of cosmetics including caffeine (such as make-up, wrinkle cream, and shampoos) is largely growing claiming some potential health benefits of caffeine. It is much surprising that only two of the products available on the market (Vichy Destock and Shower Shock) mention the actual amount of caffeine. PROLIFIC is targeting the production of novel sourced sustainably produced ingredients and cosmetic products. Although, there are current examples of raw plant materials containing compounds of interest (*e.g.* antioxidants, anti-microbial), the production of bioactive compounds with similar chemistries from novel sustainable sources can expand the market. Actually proteins originating from dry legumes are not that much used in cosmetic formulations yet they could have interesting functional properties (*e.g.* for emulsion preparation).

[51] Bastide N.M. et al. (2011). Cancer Prevention Research (Philadelphia), 4 (2), 177-184.
[52] Brewer M.S. (2011). Comprehensive Reviews in Food Science and Food Safety, 10: 221- 247.
[53] Corpet D.E. (2011). Meat Science, 89, 310-316.
[54] Calle E.E. & Kaaks R. (2004). Nature Reviews. Cancer, 4 (8), 579-591.
[55] Heckeer K. et al. (1998). Journal of the Science of Food and Agriculture, 77, 179-183.
[56] Wood P. et al. (1994). British Journal Nutrition, 72, 731-743.
[57] Weiss J. et al. (2010). Meat Science, 86, 196-213.
[58] Piñero M.P. et al (2008). Meat Science, 80, 675-680
[59] http://www.sustainablecosmeticssummit.com/Europe
[60] https://sharingbeautywithall.loreal.com/innovating/respecting-biodiversity/guaranteeing-sustainable-sourcing-our-renewable-raw-materials

PROLIFIC technologies could lead to the extraction and increased use of new protein sources from legumes, fungi and coffee. Legume proteins will facilitate emulsion droplet formation similarly as whey protein but being much more effective. PROLIFIC will produce protein hydrolysates by tailoring the functionality of peptides with selected proteolytic enzymes. Some hydrolysates will be used to produce acyl peptides and acyl amino acids for specific technological functionalities such as emulsifiers. The PROLIFIC fibre, chitosan and β-glucans ingredients will have excellent film forming, moisturising and thickening properties and will be tested for full compatibility with other ingredients. The creams and toothpaste will comply with the pH value usual in the cosmetics industry. The polyphenol, carotenoid and phytosterol ingredients of legume, fungi and coffee extracts will be tested for functional activities aimed at producing anti-age, skin whitening or sunscreen cosmetic products. These properties will be tested with in vitro tests and included in skincare formulations based on economic feasibility criteria.

5) State of the art regarding bio-sourced compounds and fractions in polymer and composite formulations for packaging

Petrochemical-based plastics, such as polyethylene terephthalate (PET), polyvinylchloride (PVC), polyethylene (PE), polypropylene (PP), polystyrene (PS) and polyamide (PA) are extensively used in packaging field because of their large availability, low cost, good mechanical performance, and heat saleability. In addition, these materials are a good barrier to oxygen, carbon dioxide, anhydride and aroma compounds. However, nowadays there is high and increasing demand for packaging made from bioplastics. In 2016, global production capacities of bioplastics amounted to about 4.2 MT with almost 40% (1.6 MT) destined for the packaging market – the biggest market segment within the bioplastics industry[61]. Rigid bioplastics applications are available for cosmetics packaging of creams and lipsticks as well as beverage bottles and many more. Materials such as PLA, bio-PE or bio-PET are mainly used. However, the market requires not only bio-based and biodegradable or recyclable materials, but also plastics with improved functionalities targeted at the specific applications.

PROLIFIC project will address this market request in four specific applications: i) anti-fungal and anti-mites coating for dry-cured meat products, ii) flexible biodegradable packaging for raw and fermented meat products; iii) biodegradable and antioxidant rigid cosmetic packaging, iv) biodegradable and antioxidant coffee capsules.

i) and ii) Anti-fungal and anti-mites coating for dry-cured meat products and flexible biodegradable packaging for raw and fermented meat products

Dry-cured meat products are known as a possible source for human exposure to ochratoxin A (OTA), a mycotoxin which has been classified as a possible carcinogen to humans (Group 2B)[62]. Due to its toxicity, the European Commission[63] established maximum admissible levels for OTA in several foodstuffs. The environmental conditions throughout the ripening time favour fungal growth on the dry-cured ham surface, such as *Penicillium nordicum* producer of OTA[64]. In recent years, to face this problem, "biocontrol" strategies were used together with the control of curing environmental conditions[65]. A different approach to extend the safety of the ham surface could be the use of edible coatings, based for example on chitosan, which has not been well explored for meat products[66]. Chitosan is a cationic polysaccharide deriving from the partial deacetylation of chitin and it is of animal (crustacean shells) or vegetal origin (fungal mycelium). In general, chitosan is characterised by a lack of toxicity, biodegradability, film-forming capacity, anti-fungal and antioxidant properties and good barrier properties of chitosan films to oxygen[67]. For these reasons, chitosan is largely used in many applications such as coatings to fruit and vegetables[68], meat[69], cheese[70] and fish[71]. Chitosan anti-fungal properties were largely investigated alone and in combination with other antioxidants and anti-fungal substances, such as essential oils[44], or with other film forming materials such as proteins[72].

We will optimize formulations based on chitosan for dry-cured meat coating by means of dipping technique to obtain antifungal and anti-mites activities. Moreover, chitosan will be applied as a coating on a PLA film in order to produce flexible packaging with biodegradability and antioxidant functionalities to protect meat products.

[61] www.european-bioplastics.com
[62] IARC (1993). IARC monographs on the evaluation of the carcinogenic risk to humans (Vol. 6). Lyon, France: IARC Press. 435–439
[63] COMMISSION REGULATION (EC) No 1881/2006 of 19 December 2006 setting maximum levels for certain contaminants in foodstuffs
[64] Rodriguez A. et al. (2012). Meat Science, 92, 728-734.
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[69] L. Higueras et al. (2014). International Journal of Food Microbiology, 188, 53-59
[70] Y. Zhong et al. (2014). LWT-Food Science and Technology, 56, 1-8
[71] M.Y. Arancibia et al. (2015). Food Control, 54, 259-266
[72] C. Valenzuela et al. (2015). Electronic Journal of Biotechnology, 18, 406-411

iii) Biodegradable and antioxidant rigid cosmetic packaging

In cosmetics many companies (*e.g.* L'Oreal), are spreading marketing messages on the use of a sustainable and responsible packaging for their products. However, biodegradable jars or bottles have not yet been introduced into the market. Moreover, packaging must be able to preserve properties of a cosmetic or personal care product, avoiding deterioration before the expiration date, and protecting products from microbial infections. Therefore, an added value for the cosmetic packaging is the capability of increasing the shelf-life of the products. Antioxidant molecules can be used in the polymer formulation to address these problems. However, several challenges still need to be solved, *i.e.* the need of using green molecules, the migration and release of molecules on the packaging surface, and the deterioration of the active molecules during polymer processing at high temperature.

We will develop innovative packaging for cosmetic products using PHA or PLA as biodegradable matrix. The use of biodegradable and bio-based polymer matrixes will be an added value versus the petrochemical-based polymers that are not bio recyclable. Polymer crystallinity, structural conformation, and molecular weight[73], will be analysed in order to assess polymer degradability. Bioactive molecules (polyphenols, carotenoids, phytosterols and caffeine) will be added to the matrix in order to increase the shelf life of the product. To avoid migration of excessive plasticizers with consequent changes in mechanical properties with ageing and to protect molecules against thermal degradation, layered double hydroxides (LDH) modified with bio-active molecules extracted in WP2, will be produced. Polymeric blends will be developed, using the residues rich in fibres of the extraction processes, for low cost applications needing reinforcement in mechanical properties and good composting rate. The presence of natural fillers or fibres will promote significantly the disintegration rate.

iv) Biodegradable and antioxidant coffee capsules

A number of companies are interested to develop fully bio-sourced and biodegradable coffee pods; fully compostable coffee capsules have been recently launched (2017) by Lavazza, Caffè Vergnano, Coop. Polymers for coffee capsules must preserve quality of coffee powder and must have a proper resistance to high temperatures used during coffee brewing.

We will optimize polymers for the production of fully bio-based and biodegradable coffee capsules and a new functionality, antioxidant property, will be imparted to the capsules. Antioxidant molecules (e.g. polyphenols and carotenoids) will be used in the polymer formulation and encapsulated in LDH inorganic clays to avoid migration of molecules due to the hot water passing in the capsules, in order to achieve resistance to temperature.

6) State of the art regarding animal feed containing bio-sourced additives

The demand for animal protein for human consumption is rising globally at an unprecedented rate. More than 80% of the protein (>40 MT of crop) required for livestock rearing in the European Union is imported from non-EU countries, much of it from non-sustainable and environmentally damaging sources, *e.g.* soy and fishmeal[74]. The availability of food for human consumption at the global level is heavily impinged upon by the large demands that livestock production places on land and water use. Approximately 6 kg of plant proteins is needed to produce 1 kg of animal protein. The global population currently stands at 6.5 billion people and is estimated to reach 9.5 billions by 2050 and emphasises the need to address how we rear livestock for human consumption. Further on, modern animal production practices are associated with regular use of antimicrobials to maintain health and productivity. The development of antimicrobial resistance in veterinary medicine causes therapy failure in animals, lowering of productivity, loss of animals, decreased animal welfare and increased costs. This urges the need for reliable feeding practices, minimizing antimicrobial use (and consequent resistance).

We will subject protein fractions to different functional tests specific for the production of novel pigs and poultry feed products. The most promising compounds will be added to feed products and benchmarked against existing soy-protein products using in vivo animal feed tests. We will also go beyond the state of the art in searching for natural bioactive compounds coping with a combined mode of action, based on a good combination of anti/pro-microbial potential, agglutination potential and anti-oxidative potential in order to reduce the use of antibiotics in animal feed.

Overall PROLIFIC Innovation potential

Tangible innovation is an essential component of the PROLIFIC project, which aims at improving extraction/modification technologies for sustainable use of three agro-industrial feedstocks and to include the recovered compounds and fractions in the formulation of product prototypes in food, cosmetic, packaging and feed sectors. The innovation cycle in the PROLIFIC project is straightforward as driven by professionals of the industrial

[73] R. Pantani, A. Sorrentino (2013). Polymer Degradation and Stability, 98: 1089–1109.
[74] http://www.euractiv.com/cap/meps-want-protein-deficit-eu-livestock-news-502925

sector. Technologies and products are simply developed with and for our SMEs and large companies, which obviously aim to increase their sales benefits by adapting their portfolio of products to address the needs of their customers and deploying early stage marketing initiatives.
The entire R&D&I efforts of the PROLIFIC project are exclusively centred around industrial partners, which have expressed needs for protein, peptides, amino-acids, fibres, chitosan and other bioactive molecules for specific product formulations and/or are seeking for biochemical solutions to increase/improve functionalities of their products. The PROLIFIC consortium strongly believes that innovation, more particularly in the field of biomass economy, must be driven by end-consumers that have clear needs in order to make a significant leap. The privileged relationship of our industrial end-users with their customers and detailed analysis of food, cosmetic, feed and packaging markets clearly point out to bright perspectives related to functional healthy food and emerging dietary trends (*e.g.* vegan diet), cosmetics with vegetal bioactive ingredients, feed with natural antimicrobials and other anti-agglutination and anti-oxidative properties, as well as readily biodegradable bio-based and active packaging. Starting from market needs and growth potential for some products, the PROLIFIC consortium has sought for underexploited sources of biomass, which contain significant amounts of the compounds and fractions necessary to procure the desired functionalities to the products that will be prototyped (Table 5). Feedback loops from the end-consumers to the industrial end-users formulating at least 16 prototypes as well as control of quality, safety and regulatory issues by the industrial end-users will constitute the main mechanisms to pull the innovation efforts of the actors of these value chains in the same direction.

Table 5. PROLIFIC's products catalogue in different market sectors.

Market	Type of prototype	Customer needs & expectations
Food	Cereal-based (bread, buns, pasta, breakfast cereals). Min. of 4 prototypes	High nutritional value and healthy food
	High-protein content vegetarian/vegan products. Min of 2 prototypes	Demand for vegetarian/vegan food with equilibrated protein content
	Meat (hamburgers, cooked and/or fermented sausages). Min of 2. prototypes	High nutritional value and healthy food
Cosmetic	Cream (with sunscreen and/or whitening and/or anti-aging properties). 1 prototype	Cosmetic with enhanced efficacy made by using sustainable actives
	Toothpaste (with anti-microbial action). 1 prototype	Toothpaste with long lasting freshness and anti-microbial protection
Packaging	Chitosan-based coating for dry-cured meat. 1 prototype	Innovative and green way to preserve meat properties against fungal and mites contamination.
	Biodegradable flexible packaging for meat products (chitosan+PLA). 1 prototype	Sustainable and biodegradable packaging, active in preserving food properties
	Rigid biodegradable and antioxidant packaging for cosmetic. 1 prototype	Bio-based and biodegradable packaging, with a low environmental impact
	Biodegradable and antioxidant coffee capsules. 1 prototype	Reduction of environmental impact, longer product shelf life and integrity
Feed	Feed for pigs. 1 prototype	Improved functional and/or nutritional characteristics
	Feed for pigs. 1 prototype	Improved functional and/or nutritional characteristics

Besides the production of these prototypes that will create clear business opportunities for SB, SSICA, COSM, ILLY and NS, PROLIFIC partners expect major innovative advances in:

- Eco-efficient extraction and conversion/stabilisation techniques with increased yields, which might be adaptable to other biomass streams. In general, the technologies start at TRL 3, making it realistic to reach up to TRL 5 (or TRL6-8 in case of food and cosmetic prototyping) within a 4 years project (UNIPR, CELAB, IRIS, IGV)
- Compounds and fractions, which might find applications in other industrial sectors (IGV, SSICA, FTEN)
- Creation of new services to advise on the best options to optimise biomass valorisation (BBEPP, IGV, SSICA, IRIS, CELAB, FHNW, UNIBO, FTEN)
- Development of novel business models for agro-industrial biomass generators (CONS, PLEUR, ILLY)
- Competitive advantage for ILLY by penetrating the market of coffee capsules with readily biodegradable and antioxidant products and benefits in terms of corporate image.

A strong Intellectual Property Rights management as well as careful patent anteriority search warranting a freedom to operate are essential to the successful commercial exploitation of the results. As mentioned in the section 2 (also in section 3, WP7) the innovation management team of the PROLIFIC project will be constituted of experienced persons from the industrial and technology transfer sector. Together with our benefit-oriented companies, the team will warranty that adequate decisions are made about keeping confidential knowhow or patenting products and processes. Acknowledging the fact that financial resources can also be a limit to innovation, the innovation management team will also offer support for the preparation of convincing business plan canvas and more mature drafts to the partners, which plan to pitch their products and technologies in front of business angels to raise funds (see WP7).

The innovation potential of the PROLIFIC project can be summarized in very few words: customer needs-oriented prototypes, industry-driven technological development, professional innovation management, early stage marketing and accessible financial tools.

2 Impact

2.1 Expected impacts

The objectives of PROLIFIC are integrally aligned with the scope of the topic and adequately address specific challenges related to the valorisation of untapped biomass streams.

As stated in previous sections, food, cosmetic, packaging and feed sectors are facing the reduction of conventional sources of active ingredients and new environmental-friendly demands from the final users. EU Commission 2017 publication on comprehensive EU protein sector[75] reveals a remarkable decrease in protein crop production in the EU in the past ten years, contributing to the already existing alarming dependence of the Union on the imports of protein crops, used mainly for food and feed sectors. In terms of potential market demand in EU, the protein demand for food is mainly covered by foreign import with 37 MT/year (68% of total supply) of imported proteins[76]. The market of bulk chemicals has also a substantial demand in EU (2.4 MT/year) including biopolymers, bio lubricants, bio-solvents and surfactants[77]. For the projected increasing world population (up to about 9.5 billion by 2050) new trends in human nutrition are observed, *e.g.* utilizing protein-rich plant resources, proteins from algae or insects[78],[79],[80], but also non-soy[81] and gluten-free protein products[82]. Meanwhile the worldwide demand for animal-derived protein is also expected to double by 2050[83]. The market forecast for protein and other bioactive ingredients has been reported to grow at a compound annual rate of 6.0% reaching 58.5 billion USD by 2022[84]. The rising market for nutritional food focused on vegetal protein ingredients is ascribable to the major trend among customers to reduce meat consumption for health and sustainability reasons. Especially the health and fitness oriented consumer groups have an above-average willingness to pay for special nutritional benefits. Currently, vegetal protein products including those with a high protein content are mainly based on soy, dairy or wheat proteins, and are very costly with prices between 17-33 €/Kg[85]. Newer products based on pea, lentils on other leguminous seeds are rapidly growing. Besides, European consumers are looking for non-allergenic, GMO-free, gluten-free, nutritious and safe alternatives. Data on changes in consumer's culinary preferences reveal that the market for non-dairy products is dramatically increasing and that meat alternatives sales are expected to reach 5.9 billion USD by 2022 and more than 1/3 of consumers are open to plant-based products[86]. Mintel, the global leader in market intelligence, included animal product alternatives as a major pillar in its Global Food and Drink Trends 2017 report[87], reporting that the 3 top-selling groups of goods will be vegetarian, vegan meat and milk alternatives, and the breakfast product segment, with a significant slowdown in the growth of demand for meat in Western Europe.

[75] Overview of protein crops – a useful tool in CAP (Common Agricultural Policy) post 2020 discussions
[76] FAOSTAT, 2013
[77] Eurostat, 2010
[78] United Nations Department of Economic and Social Affairs/Population Division and World Population Prospects, 2012
[79] Spiegel M. et al. (2013). Comprehensive Reviews in Food Science and Food Safety 12.6, 662-678.
[80] Petrusán J.I. et al. (2016). Current Topics in Peptide & Protein Research, 17, 1-19
[81] Christopher Shanahan, Global Program Manager – Food&Agriculture Practices, Frost&Sullivan (2014) Size, Share, Sources and Segments: Analysis and Forecasts for Value-added Protein Ingredients, Protein Trends& Technology Seminar, Illinois, USA, April 8-9 2014
[82] Melinda Dennis, LDN- Combining Vegetarian, Vegan and Gluten-Free Diets, RD Resources for Consumers, Academy of Nutrition and Dietetics. 2014
[83] FAO Report 2013
[84] MARKETS AND MARKETS (2016): Meat Substitutes Market by Type (Tofu & Tofu Ingredients, Tempeh, Textured Vegetable Protein, Seitan, Quorn), Source (Soy-based, Wheat-based, Mycoprotein), Category (Frozen, Refrigerated), and Region - Global Forecast to 2022. http://www.digitaljournal.com/pr/3226886
[85] Robert Joyce Commercial Business Consultant, Stewart-Peterson Inc.: Soy & Dairy Market Volatility-Its Impact on Protein Prices and Strategies for Managing Financial Risk, Protein Trends& Technologies Seminar, Illinois USA, 8-9 April 2014
[86] Innova Market Insight: www.innovadatabase.com
[87] Global-food-and-drink-trends-2017.pdf at http://www.mintel.com/global-food-and-drink-trends

As previously stated, the consumption of **meat** products in most EU countries is stagnating, because the consumers are becoming more critical with regard to health and safety aspects of foods in general. In particular, in the last years the consumption of the minced meat products *(i.e.* fermented and cooked sausages) strongly decreased (*e.g.* -7.6% between 2015 and 2016 in Italy[88]) due to their unbalanced nutritional profile (WHO recommendations). On the contrary, the meat products market segment indicated as 'light and healthy' is showing considerable growth during the last 10-15 years, and the design of new minced meat products, enriched in phytochemicals and therefore with a better nutritional profile, is contributing to recover the loss of the EU general market share.

PROLIFIC is not only following sustainability aspects of global emerging markets[89], but also addresses the present European food policy on the provision of safe, nutritious, high quality and affordable food to European consumers.

Consumers are also increasingly demanding a **food and cosmetic packaging** sustainable, at low environmental impact, and also able to preserve the quality and increase shelf life of the products. In this perspective, to meet this request, bio-based and biodegradable/recyclable plastics and green additives can play an important role. Even if today bioplastics represent only about 1% of the about 300 MT of plastic produced annually, according to the latest market data compiled by European Bioplastics, global production capacity of bioplastics is predicted to grow by 50% in the medium term up to 6.1 MT in 2021. Moreover, the demand of bioplastics grows up together with the need for more sophisticated materials, applications, and products. Concerning coating for meat products, greater attention is focused today on alternative packaging strategies able to control frequent moulds and mites contaminations. These would reduce at the same time the environmental impact of synthetic substances actually used against contaminations as well as guarantee safer dry-cured meat products. Moreover, the use of a new bio-based packaging preserving meat products from oxidation phenomena could provide a positive impact for consumers in terms of nutritional profile.

In the fields of bioplastic and packaging market, **PROLIFIC** activities will contribute firstly to meet the increased request for bio-based, active, and biodegradable molecules and polymers to be used for polymer formulations and applications. Moreover, PROLIFIC will contribute to introduce and validate the use of bio-polymers and green additives in the packaging applications, facilitating the growth of bioplastics sector thanks to their advanced technical properties, potential cost reduction, high consumer acceptance and boosting climate change mitigation.

The EU demand of protein sources for animal feeding is projected to increase from 49 MT in 2016 up to 55 MT in 2026[90] and the actual market value for animal feeding proteins (mainly from soy) ranges from 0.5-1 €/kg. Instead the functional feed ingredients market value varies from 10 to 500 €/kg (depending on function, efficacy and concentration), with a Worldwide market value (estimated by NS partner) of 15.4 billion USD of which 3.5 billion USD in Europe only[91]. The expected market growth was estimated by NS to be around 3-5%/year. Protein fractions develop during PROLIFIC project, are benchmarked to be able to compete with actual conventional commodity protein sources (such as soy, Table 4). The addition of bioactive compounds extracted from novel biomass sources (*e.g.* polyphenols and carotenoids), specifically selected with the aim of reducing the dependency of antibiotics will be valuable for the market of functional feed ingredients.

As PROLIFIC aims to become the leading solutions provider for ingredients/additives alternatives or complementary to those currently present on the market, the range of final protein ingredients will include proteins, peptides, amino acids, fibres, chitosan, β-glucans, polyphenols, carotenoids, phytosterols and caffeine (Table 4). The development of prototype products including one or more of the previous fractions will contribute to cross-link four different market sectors: food, cosmetic, packaging and feed (Table 5). Local growth and local residue valorisation combined with competitive and cost-effective extraction methods will contribute to decrease imports of these components and increase the competitiveness of local resources. Alternative solutions, from agri-food residues to traditional crops with higher performance in terms of concentration of nutritional elements and efficient extraction procedures will be developed under the PROLIFIC project. Herewith we address one of the main priorities of the BBI JU Work Program for 2016: ''***Sufficient and sustainable biomass feedstock supply at affordable price incorporating the feedstock suppliers as partners in the value chains***'' and more specifically with the ''***Development of biorefinery technologies and processes to increase the competitive position of the bio-based industries in Europe***''.

<u>Contribution to the expected impacts set out in the BBI work programme, under the relevant topic</u>

PROLIFIC project will address the expected impacts as requested in the BBI-R4 call, as follows:

Contribute to KPI 1: create at least 1 new cross-sector interconnection in bio-based economy clusters

We expect creating new opportunities for knowledge exchange and business purposes within the PROLIFIC project. In fact, several new cross-sectorial interactions between companies will be created. For instance none of the three

[88] Annual Report 2016 of ASSICA
[89] Babault N et al (2015). Journal of the International Society of Sports Nutrition, 12, 3
[90] https://ec.europa.eu/agriculture/sites/agriculture/files/markets-and-prices/medium-term-outlook/2016/2016-fullrep_en.pdf
[91] Animal Feed Additives: a global strategy Business Report June 2007" report, published and distributed by Global Industry Analysts, Inc., www.strategyR.com.

providers of biomass previously cooperated with industrial end-users of compounds and fractions, with the only exception of CONS and SSICA. In other words, three biomass suppliers (CONS, PLEUR, ILLY) from agro-food sector will establish new cooperation with the cosmetic (COSM) and packaging (FTEN) sectors. Besides several new interactions between academia and private sectors, *e.g.* FHNW (strong focus environment and resources) and IGV (food focus) are expected to open new horizons in terms of possible research synergies such as in the field of trace and speciation analyses of heavy metals. Also the creation of new interactions between academics are worth mentioning, *e.g.* FHNW and UNIPR (food focus) that will synergistically work on the implementation of protease-based hydrolytic process.

2) Contribute to KPI 2: set the basis for at least 1 new bio-based value chain

The PROLIFIC project will set the basis for at least five completely new bio-based value chain and in particular hyper-proteic allergen-free vegan food, bioactive and biodegradable food packaging, bioactive and biodegradable cosmetic packaging, bioactive coffee capsules, functional feed having novel anti-microbial properties. All PROLIFIC new value chain will be tested and validated at prototype level by industrial developers and end users. LCA and CEA assessments will proof they are economically viable and fulfil all relevant sustainability criteria. In particular the use of legumes, fungi and coffee protein-enriched extracts will allow to produce vegan food products with a high nutritional value having being at the same time allergen-free if compared with direct competitors based on soy-proteins, wheat proteins or egg-proteins.

Innovative bioactive and biodegradable food and cosmetic packaging will be developed by using the biodegradable polymers (PHA, PLA) to which bioactive molecules (polyphenols, carotenoids, phytosterols and caffeine) will be added in order to increase the shelf-life of the packed food or cosmetic product. The use of biodegradable and bio-based polymer matrixes will be an added value versus the petrochemical-based polymers, not recyclable, utilised nowadays in particular for cosmetic packaging. Fully bio-based, biodegradable and bioactive coffee capsules will be produced by using biodegradable polymers materials added with PROLIFIC antioxidant extracts. The new functionality of the capsules will lead to a higher integrity and shelf-life of the coffee product. Finally, PROLIFIC will lead to the obtainment of animal feeding products added with functional natural anti-microbial ingredients aiming for a substantial reduction of the use of antibiotics in animal feed.

3) Contribute to KPI 5: set the basis for at least 2 different ingredients or additives with purity levels that enable their use in different application fields, such as the food, feed, flavouring, fragrance, cosmetics, chemicals, textile, nutraceutical and pharmaceutical industries

The PROLIFIC consortium will isolate several compounds and fractions, namely proteins, functional peptides, amino acids, fibres, chitosan, β-glucans, carotenoids, polyphenols, phytosterols and caffeine from legume, fungi and coffee residues. These new compounds and fractions, extracted from the streams, with interesting properties will allow the development of improved new final product prototypes with novel features and the replacement of many chemicals in their formulations. The new bio-based ingredients will substitute traditional chemical compounds in final prototype formulations for food, cosmetic, packaging and feed sectors. PROLIFIC will extract these compounds and fractions and use them in the formulation of at least 8 different food (cereal-based, vegetarian/vegan, meat), 2 cosmetic, 4 packaging and 2 feed prototype products (WP5). Concerning the **food** products, proteins, peptides, amino acids, fibres, chitosan, polyphenols, carotenoids and caffeine, will be used in the production of different cereal-based, vegetarian/vegan and meat products. The main properties of the compounds and fractions to be included in **cosmetics** are those with film formation, UV filtration, whitening, antimicrobial, anti-ageing and antioxidant. Concerning the **packaging** sector, active (anti-fungal and antioxidant) packaging exploiting polymers (chitosan) and biomolecules extracted from the selected residues. In **feed** sector, the most important ingredients will be high-proteic and carotenoid-rich fractions for the substitution of traditional sources (vegetable crops as soy, rice or wheat) and bioactive molecules (*e.g.* polyphenols) having antimicrobial properties so to substitute current used products and improving animal health. The purity, functionality and compliance with food, cosmetic, packaging and feed market standards and regulations will be evaluated in WP4, WP5 and WP6. The recovered compounds and fractions will be benchmarked vs. commercially available similar products in terms of quality and price (see Table 4).

4) Set the basis for further development of new or optimised value chains based on residual streams from the food or feed industry, agricultural activities or biorefineries

The PROLIFIC project will concretely address the BBI impact objective of establishing of solid basis for the further development or optimised value chains based on residual streams from agro-food industry by producing and distributing a total of 16 prototype products (see Table 5). Experimental results, technological developments as well as accompanying analyses (LCA, CEA, industrial environment, regulatory and safety issues, sensory and marketing tests) generated up to the production of the final prototypes constitute the best basis for the development of new or optimized value chains. Reaching the product prototype stage will constitute a pre-industrial validation of final consumer products for our industrial end-user partners from the food, cosmetic, packaging and feed sectors. The latter are in fact the most suited actors of the envisaged value chains as to the promotion and real implementation of

the optimal valorisation of up to date still underexploited legume, fungi and coffee feedstocks. In fact, the industrial end users will define the critical needs and requirements related to production processes of high added-value products from agro-industrial residues and they will accordingly instruct research entities and technological companies. Final food products will be validated by the partners IGV, SB and SSICA. COSM and IGV will be the industrial partners in charge of validation of the final cosmetic products. The new functionalities imparted to the end-packaging products will be validated by SSICA, ILLY, FTEN, and COSM industrial partners. Feed market will be addressed by the development of piglet and poultry feed with corresponding validation by the industrial partner NS.

5) Overall reduction of at least 10 % in the carbon footprint of the considered bio-based operation compared with the state-of-the-art (shown by an LCA taken up in one of the work packages).

During PROLIFIC, sufficient amounts of target compounds for application tests and production of several prototypes will be produced and upscaling of the extraction/transformation processes will be performed (Task 5.1). This will allow evaluating the robustness, effectiveness, real cost-reduction and applicability of the designed processes. The methodology and technological approaches in the PROLIFIC project represent several opportunities to decrease the carbon footprint associated to the valorisation of underexploited biomass streams. The combination of selected streams, extraction/conversion procedures and final product processing techniques will not only be useful to optimize the processes and improve product quality but also add to the economic and environmental viability of the targeted value chains. The achievement of the 10% decrease value of the carbon footprint will be achieved by adequate technological choices and verified by means of accompanying LCA (WP6). A good example is the use of planetary roller extrusion for production of cereal-based, vegetarian/vegan and feed products, which will allow an increase in process and energy effectiveness versus state-of-the-art single- and double screw extruders. Furthermore, the potential to include an improved supercritical CO_2 step in the biorefinery approach as a "green" technology for the extraction, purification and formulation of ingredients from the selected feedstocks will be demonstrated during the PROLIFIC project. At the drying stage, the use of horizontal spray drying will reduce costs of drying with ca. 40%, meaning an overall impact of ca. 20% on products final costs. Concerning transformation reactions, enzymes reduce the environmental footprint of manufacturing by reducing the consumption of energy and chemicals and concomitant generation of waste. For instance the proteolytic treatment of hide in the leather industry enables the saving of 97 kg CO_2 per ton hide as compared to classic chemical methods. Furthermore the operation of process using immobilized biocatalysts in reactors retaining the immobilized enzymes enables to save at least 10 times the amount of biocatalyst compared to process based on soluble enzymes. The protection by organo-silane is expected to drastically increase this factor. The technological procedures will also allow a reduction of the amount of solvents used, making use whenever possible of aqueous extraction processes. In addition, efforts will be made to reduce costs via developing processes (where possible) on wet biomass, as drying step requires high energetic costs.

With the development of PROLIFIC products and technologies and a flexible biorefinery cascading approach, the biomass utilization efficiency will be increased by 20-50%, which will be correlated with an estimated 15% carbon-footprint reduction achieving both effectiveness and cost-reduction of the processes.

Contribution to business and innovation

Innovations in products and production processes, together with sustainability and efficiency at competitive costs are key factors for the EC to extend local products, covering the whole value chain, in local markets, reducing imports and promoting exports to new markets.

PROLIFIC will contribute to the development of the EC strategy for this topic, since it will develop competitive and cost-effective alternatives to current traditional resources such as the protein, peptides amino acids and other bioactive molecules used for applications in food, cosmetics, packaging and feed sectors. Integration of new extraction and modification methods, together with research on new compounds and applications thereof, and the collaboration in all steps of the value chain, from agri-food residues to final industrial end-users, will derive in **novel competitive products** of high quality at competitive prices, reducing the current dependency from imports outside Europe.

The project offers to the companies involved new possibilities to **improve their innovation capacities** and **expand their business models**. PROLIFIC products, obtained from new and optimized extraction processes, from selected residues, will be introduced into the market through final applications by industrial end users. Traditional compounds, for food, feed and cosmetics will be substituted using biomass feedstock residues generated in Europe to produce new compounds or fractions of higher performance. First prototypes developed under the project will be an initial step for the launch of further applications in the sectors involved as well as other application in the frame of future additional cross-sectorial interactions. The impact will not be restricted to the participating consortium since the recovered compounds and fractions will be used for further applications to different related subsectors thanks to significant efforts in dissemination communication and commercial exploitation. All levels of the value chains will

benefit from the innovations created by the PROLIFIC consortium and will be a key factor to increase the competitiveness of the industrial partners involved in the short term.

- **Residue "producer" companies** (CONS, PLEUR, ILLY), will have new opportunities to valorise their feedstocks and increase future revenues from the sale of residual biomasses. In addition, these residue producer companies will decrease residues disposal costs.
- **Formulation, extraction and evaluation partners** (FHNW, UNIPR, UNIBO, IRIS, SSICA, CELAB, COSM, NS, IGV, SB, FTEN) will obtain new product lines and new optimized extraction processes to offer competitive solutions for the sectors involved.
- **Industrial end-users** involved in PROLIFIC (ILLY, SB, IGV, COSM, FTEN) will be pioneer in producing these innovative products. Their experience and capacities will be having first market penetration.
- FHNW will acquire valuable data for its consulting activities on biomass valorisation by performing complete **Life Cycle Analysis and CEA** along all steps of the value chain of the products developed.

Real collaboration is one of the most challenging organizational values that are being developed in global value chains. Collaboration must be based on trust, information and risk sharing and the involvement of all the participants from research and conceptual design to his grave. PROLIFIC will involve all actors mentioned for the success development of future marketable products. The collaboration of different RTD performers (FHNW, UNIBO, UNIPR, SSICA) in the project will provide competent technological support to the consortium and bring into application last innovations in all technological steps following rigorous guidelines laid down by industrial end-users. The integration of different industry partners, RTD centres, technologic SME companies for formulation and extraction processes, producers and industrial end users (SMEs and large companies) will compose a powerful consortium. The wide experience of TALOS in communication will also be crucial to intensive exchanges between the participants, creating an effective and favourable environment for the exchange of know-how in all relevant areas. The ambition of PROLIFIC is to **convert new knowledge into process innovation and mainly new prototype products** by building on existing knowledge and experience in key areas. The dissemination and communication strategies, to be undertaken during the project under the lead of TALOS (see Section 2.2), will also ensure that the knowledge developed during the project will be shared with the wider community and a large number of stakeholders, provided by INNOV, in order to promote subsequent exploitation of the technologies.

Collaborations with other related projects (see Section 1.3) and direct contact with other interested entities will constitute additional opportunities to receive external inputs and feedback. The participation of the stakeholder advisory board - a wide experts panel covering the bio-based value chains (association, public entities, final users, *etc.*) - will be an asset in realising the aforementioned ambition of the PROLIFIC project.

Contribution to other environmental and socially important impacts

A key contribution of the PROLIFIC project is to achieve bio-based industrial growth and competitiveness of our companies without exerting negative environmental impact. We will show that their economic development goals do not affect climate change, energy insecurity and resource scarcity and that outcomes will be in line with conclusions made with Green Economy Report[92]. In the PROLIFIC project, food, cosmetic, packaging and feed sectors actively drive the innovation cycle and benefit from technological advancements and new knowledge to implement sustainable innovation that will lead to eco-friendly production and processing and have impact in terms of revenues and job creation. The new products developed using compounds and ingredients will have positive impact on well-being and health. Healthy cereal-based, vegan/vegetarian products and meat products rich in proteins or in bioactive compounds, as well as the production of more green and active cosmetics, biodegradable and bioactive packaging and feeds having less chemicals and more functional characteristics (*e.g.* antioxidant and anti-microbial), are pertinent examples. The PROLIFIC project will contribute to reach the objective of increasing sustainable food production (including agricultural feedstocks and valorisation of residues/wastes) in Europe (roadmap to resource efficient Europe) and of mitigating environmental pollution resulting in climate change, land degradation and biodiversity decrease. The outputs of the PROLIFIC project will be disseminated all over Europe by the partners and by members of the Stakeholder Advisory Board as well as targeted groups of stakeholders.

The utilisation of the agro-industrial processing residues/co-streams using flexible cascades of processes enhances the financial returns to the agri-food industry, thus improving its competitiveness. This improvement in industrial processing efficiency and economic return moves down into local communities, delivering valuable commodities and dividends that are both financially and environmentally tangible at local, regional and pan-European levels. This has the inevitable effect of enhancing Europe's position as a leader in sustainable agro-industrial processing systems, adding to brand value and market positioning for PROLIFIC-like flexible **biorefinery processes.**

[92] Michael Hermann (2009). The impact of the European Novel Food Regulation on trade and food innovation based on traditional plant foods from developing countries. Food Policy 34 (2009) 499–507

Barriers/obstacles or framework conditions that may determine whether and to what extent the expected impacts will be achieved

The use of compounds or fractions obtained from legumes, fungi and coffee residues for food, cosmetic, packaging and feed applications as substitutive of *e.g.* traditional protein sources (such as cereals or animal protein) may constitute an obstacle for their introduction in marketable products. There are differences depending on the sector and the type of applications regarding technical regulations and standards related to the use of the obtained products. Social acceptance of agri-food residues for human consumption can be a market barrier in European society. Concerning food industry, the products developed within PROLIFIC **will not need** the positive assessment of the *European Food Safety Authority (EFSA)*. However, EFSA will be consulted in order to eliminate any hurdles in potential commercialization of derived products. Table 6 shows the main barriers and obstacles that PROLIFIC has to face in its R&D&I activities as well as measures to circumvent these difficulties.

Table 6. Barriers/obstacles and foreseen mitigation plan during PROLIFIC.

Issues	Barrier/obstacle	Mitigation plan
Market	**Food market** Conservative market. Acceptance of food residues extracts for human consumption	Initial and early information to potential consumers, involving public authorities. Information of direct benefits but also environmental when labelling products containing PROLIFIC compounds. Give-aways of prototypes and product evaluation questionnaires
	Cosmetic market Achieve competitive prices for substitution of traditional sources.	Make cost-effectiveness as a must in all stages of value chain, optimizing processes and resources. Green image communication, give-aways.
	Packaging market High prices of a bio-packaging with respect to petro-derived polymers	Dissemination actions towards consumers to explain the advantages of a bio-based and compostable packaging. Dissemination on the advantages of an active packaging for food and cosmetic preservation. Green image communication
	Feed market Cereals are established source of nutrients for feed, cheap and with high availability.	Active dissemination actions focused on the advantages (quality, eco-efficiency, price) for the feed industries vs traditional sources.
	Ensure availability of compounds for high demand.	Plan the industrial scaling-up in all steps of the process. Design procedures and methodologies thinking in future large scale industrial applications. Consider adding further biomass sources.
Regulations	Restrictive legislation for food, cosmetic, packaging and feed applications.	Take into account all requirements from the beginning and on each step of the value chain.
		Developing alternative composition/processing routes in each process in case a substitution or modification is required.
Financial-economic	High labour costs	Combination of partners from countries with different labour costs (lower in southern countries) will result in competitive personnel costs along the whole value chain. Pricing efforts.
	Public research rarely focuses on up-scaling and optimising production.	PROLIFIC consortium composed of 4 RTD centres and 13 industrial partners in order to reach a significant synergy that ensures R&D&I of new bio based value chains built on coffee, legume and fungi residues towards food, feed and cosmetics markets. Industrial end-users driven action with 16 concrete prototypes within four years.
	Large companies hardly invest in R&D in this field	PROLIFIC includes the participation of large companies (CONS, ILLY) for future commercialization of the results. They also participate from the definition of the project by taking into account their expertise in alternative sources or applications. Access to European Business Angels network via TALOS.

2.2 Measures to maximise impact

a) Dissemination and exploitation of results

The PROLIFIC project places significant emphasis on the exploitation and dissemination of results, which will be acquired throughout the whole duration of the project. Dissemination and communication activities of the PROLIFIC project are designed to ensure maximum impact; consortium partners will adopt a multidimensional dissemination and outreach framework where every dimension targets a particular community in a specific way. The dissemination strategy of the project is not just a scientific awareness of the innovative results, derived from the project, but also focuses on other target groups such as professional, commercial, investment, social, environmental, policy-making as well as educational and training activities (including specialised workshops). Potential stakeholders of the project include the representatives of food, cosmetic, packaging, feed, biotechnological, nutraceutical and pharmaceutical industries, agriculture, agro-industrial biomass providers, technology transfer, investors, suppliers of technological devices for extraction and modification, academic, research community, relevant on-going research projects (BBI/H2020), national and regional authorities, high-tech centres and incubators.

Involving stakeholders: The aforementioned groups are important stakeholders for the PROLIFIC project. Communication channels will be established with them during the implementation of the project. It is important to note that links between the PROLIFIC consortium and all of these stakeholders already exist but will be enriched and deepened during the implementation of the project. Specifically, stakeholders will be invited to participate on workshops in order to record their needs and opinions and take these inputs into consideration during R&D&I and marketing activities. Networking with other ongoing research projects will also be established in order to exchange best practices and organise joint dissemination events for enhanced impact (Task 7.2).

Networking with stakeholders will be one of the main axes of the dissemination activities but it will also have the purpose of actively involving the target groups in the project activities with the final aim to receive their feedback on the project results as well as to elaborate further project developments. As mentioned in Task 7.2, networking will be implemented at different levels with the following target groups: a) synergistic projects and initiatives (in order to enrich research activities and enhance dissemination via joint events), b) end-users / customers (in-order to pave the way for future exploitation and commercialisation), c) potential investors / business angels (in order to secure private investment that can enhance product development and the impact of market entry).

In order to create awareness, engage crowdsourcing activities and maximise impact, PROLIFIC consortium will employ a range of means to disseminate its concept and results. These will include an online presence, oral and poster presentations, written publications, active contributions to scientific publications, conferences and, workshops, involvement of key stakeholders, organisation of events, public discussions, links to other EU funded projects, etc. These dissemination activities will take place in accordance with the Consortium Agreement that has been signed, and protect any commercially sensitive material. Concerning the organisation of workshops, already **2 workshops** have been agreed and specified by the partners: one of them will be organised by ILLY in Trieste towards the end of the first year of the project. This workshop will include also a tour at their production plant and the workshop will be organised in the conference hall at their main production site. The main focus of this workshop will be the residual biomasses from agro-industrial processing. The second workshop will be organised by IGV in Potsdam towards the end of Year 3 of the project and will focus on products from ingredients extracted from residual biomasses and their exploitation and commercialisation potential.

The goal of dissemination is to promote and raise awareness about the project achievements, and communicate its benefits to the targeted stakeholder community. Thus, the project will create and implement a Communication and Dissemination Strategic Plan at the beginning of its timeline; key partners in the consortium will collaborate in defining and setting objectives, and during its implementation will seek the active engagement of all partners' networks and contacts. The plan will play a central role to the success of the PROLIFIC project. TALOS, as Dissemination Manager will be responsible for ensuring that the consortium carries out these activities as well as recording any achieved activity independently. This dissemination plan is a working document, that is, the plan will be updated regularly to reflect appropriate dissemination activities during the lifetime of the project.

The implementation of the dissemination plan will be based on the following approaches:

1. Identifying with precision the target audiences, mapping the project's stakeholders across Europe and preparing practical tools to approach them (mailing lists, registration to discussion groups, social web presence, information material, *etc.)*
2. Establishing the engagement tactics to be adapted to different audiences
3. Exploiting partnership networks

The following actions will be implemented and fully described in the Communication Plan (contained in D7.1 Dissemination and Communication Strategy and Action Plan) to maximise outreach and impact:

- Gathering supporting partners during the project lifetime
- Supporting partners who will contribute to the implementation and use project results
- Organising regular events for consultation (e.g. workshops, conferences and events with local stakeholders) that will feed back to the project activities, building consensus with external stakeholders throughout the project and assist Networking and Policy advice activities
- Leveraging in existing networks of people and organisations to maximise outreach and impacts
- Providing the project's portal and online web tools to build up the community
- Using social media and classic media for awareness and engagement purposes

Organisation of dissemination activities: the dissemination activities will be based upon a concrete plan for dissemination (D7.1, due in month 5) which will define clear objectives adapted to the relevant stakeholders and set up concrete protection, exploitation and dissemination strategy which will be followed during the project lifetime. To maximize the protection, exploitation and dissemination of the project results, a draft Exploitation Plan (D7.4) will be prepared by the partners and will be updated during the project implementation based on the input and feedback that the consortium will receive from the different stakeholders. The development of both D7.1 and D7.4 is important in order to actively involve stakeholders, target audience and the public to the project from the initial stages of its implementation and thus ensure excellence and the widest possible participation. Exploitation and dissemination measures will be adopted to ideally address potential end-users and optimize the use of the results that will be generated. Such measures include for example research activities, commercial exploitation activities, standardisation, skills and educational training as well as policy-making.

b) Communication activities

A variety of communication activities will be required to reach the diverse range of stakeholders which have been identified in Section 2.2 a). The communication activities described in this section will be updated in D7.1.

Website and social media: The consortium will create various public channels for the dissemination of the PROLIFIC activities and results, as well as related content. The **website** will include both public and private-access pages. A common framework will be provided through the website to unify research deliverables and content and allow comparative studies among different countries and stakeholders. The private section of the website will be used for posting of key project documentation among the consortium members. **Public social media accounts** will be created (twitter, facebook, linkedin, *etc.*) and they will be complemented by a unified social media strategy for the whole consortium that will be implemented to engage users and spread the word about the project results in a coherent manner, creating synergies with each partners' own social media channels.

Press releases: Press releases will be prepared to disseminate key project achievements. These press releases will include among others: research results, presentations at conferences, announcement of workshops/other events, achievements and future planned development activities. Press releases will be published at least every 6 months.

Printed and online material: WP7 leader will create a single graphical identity for the PROLIFIC project. Traditional dissemination material, including brochures, flyers, posters and slide-sets will be made available to all project partners. Most of the dissemination and training material will be designed in English, but ready for multilingual implementation. Translations from English to partners' languages will be managed by the partners, on the basis of need, in their respective countries.

Scientific publications in journals and academic conferences: PROLIFIC project will also follow the publication approach in order to disseminate the main results of the project as long as no confidentiality or patentability are not endangered. Many partners, especially the academic partners, are planning to present papers at conferences (poster or oral contributions) and publish papers in selected peer-reviewed journals. The major peer-reviewed journals being targeted will be indexed to Scopus and/or Web of Science, online scientific databases, which keep the maximum standards for academic publications, promote the best research practices, and provide tools to track research analysis and citation metrics long after the project lifespan. It has been already agreed among the partners that for the scientific publications the Gold open access publishing model will be used as far as possible, which means that the articles will be immediately provided in open access mode by the scientific publisher and at the same time the associated costs will be shifted from the reader to the research institution making the publication. Indicatively we will target the following journals: Food Chemistry, Journal of Agricultural and Food Chemistry, Packaging Technology and Science, Meat Science, Food Control, Bioresource Technology, New Biotechnology (Administrative Coordinator is in the editorial board). PROLIFIC partners intend to participate in the following conferences: Recent Advances in Food Analysis, EuroFoodChem, EuChemS, Conferences on Food Waste and Byproducts management, FoodOmics, ICoMST (International Conference of Meat Science and Technology), International Conference of Environmental Engineering and Management (ICEEM), European Congress on Biotechnology (ECB). The administrative

coordinator is one of the chair of the ICEEM series and scientific board member of the chair and scientific committee of ECB as well as professional fairs like Ecomondo (the administrative coordinator is member of the scientific board of this fair).

Events: A programme of specific events organised by PROLIFIC will take place across Europe, with high European impact, possibly co-located with major events in order to be able to attract more attendees.

a) **Workshops** will be organised as a side-events for major scientific conferences as well as locally based on the needs of the local stakeholders. Thematic workshops / seminars / trainings are planned in the following thematic areas: residual biomasses from agro-industrial processing (organised by ILLY) and products from ingredients extracted from residual biomasses and their exploitation potential (organised by IGV). Additional workshop specifically addressed to PhD students of the participating universities will be organized by FHNW on bioconversion processes and by IGV on innovative extrusion and expansion processes for food and feed.
b) **In-store boosts** will be organised by SB, IGV and COSM where PROLIFIC ingredients and prototypes will be distributed to customers as giveaways. The aim of these in-store boosts will be two-fold: on one hand to obtain feedback from the customers regarding their appreciation of the product (questionnaires) and on the other hand to pave the way for future commercialisation of the PROLIFIC products.
c) PROLIFIC partners will participate in major European and international **exhibitions and fairs** in order to reach the industrial community and raise awareness among them about the project results. Already identified exhibitions and fairs where PROLIFIC partners intend to participate: VeggieWorld, IBA, AnugaFoodtech, Ecomondo.
d) Partners will take part in presentations, academic **conferences** and distribution of dissemination material in external and third party events, both public events and those organised by other networks and stakeholders, as part of the project's dissemination, communication and awareness raising activities.

Scientific exchange: PROLIFIC will actively seek links and interactions with other EU-funded projects in the areas relevant to PROLIFIC. These projects will be approached with the aim to discuss sharing of disseminated materials and the possibility to develop collaborative dissemination activities such as joint workshops.

Table 7 summarizes the planned dissemination and communication activities, impact and performance indicators.

Table 7. Planned dissemination and communication activities and performance indicators.

Dissemination activity	No. of planned activities & key performance indicators	Targeted audience	Expected impact
Project website	1 website, at least 15.000 unique visits	General public, all stakeholder groups	Raising general awareness, informing interested audience about PROLIFIC
Leaflets, posters	3 versions of each (new version every 2 years) - 6.000 copies of leaflets, 500 copies of posters		
Logo	1 logo for the project		Establishing the visual identity of the project
Workshops	At least 2 workshops with at least 25 participants on each workshop	PhD and graduate students	Raising awareness on recent developments in bio-based value chains
Investment workshop	At least 1 workshop with investors	Business angels	Attract funding for development of some products
Scientific publications	At least 12 scientific publications in peer-reviewed journals	Research, academic, scientific community	Enhancing the scientific impact of the project, establishing research collaborations
Participation at scientific conferences	At least 10 participations at scientific conferences		
Participation at exhibitions and fairs	At least 30 participations at exhibitions and fairs	Industrial community	Enhancing industrial, commercial and market impact
In-store boosts	At least 10 in-store boosts organised by SB, IGV and COSM	Customers, end-users	Enhance commercialisation and exploitation potential
Newsletters	At least 8 newsletters (2 per year)	General public, all stakeholder groups	Raising awareness, informing interested audience about Prolific
Social media presence	Facebook, Twitter and LinkedIn account (at least 2.000 likes and group members via social media)		

c) Intellectual property and knowledge protection

Knowledge management and protection will be an integral part of the PROLIFIC project. Adequate decision will be made to keep confidential know-how or publish it in the form of a patent. Detailed knowledge management and protection guidelines will define the framework and practical actions for the management and protection of the generated knowledge. Standard IPR policies will be followed according to the signed consortium agreement (CA). As starting point for the project, some partners will use IPR background related with the activities to be developed. Previous identified patents (in preparation, pending or granted) and relevant background related with the contents of PROLIFIC project are precisely included in the CA that has already been signed prior to the start of the project. This document will contain the IPR and knowledge protection rules during and after project execution. The regulation of intellectual property rights will be set out in detail in the CA but PROLIFIC partners agree on the following concepts:

- Within the first six months of the project implementation, either an exploitation or a knowledge Plan for Dissemination and Use of Knowledge (PDUK) will be mutually agreed between the Parties.
- As per article 24 of the EC-GA, each Party is the exclusive owner of its "Background". The Parties (incl. their Affiliates), recognizing the impact that restrictions on access rights to Background may have on the content of the Proposal, fully agree to set up a final list of the Background to which they agree to grant access rights to other Parties and to state in this list the conditions under which they agree to grant said access rights to others Parties in accordance with the terms of the Grant Agreement (GA); this list is part of the CA that has been signed, and shall be revised during the execution of the Project in case of major changes in Work Packages.
- According to article 26 of the EC-GA Results", shall mean any (tangible or intangible) output of the action such as data, knowledge or information that is generated in the action, as well as any rights attached to it, including intellectual property rights (hereinafter, the "**Results**") will be the property of the Party that generates it.
- Therefore, each Party will be the exclusive owner of what it develops on its own (without the use of another Party's Background).
- In the event that several Parties have jointly developed Results (i.e. where the separate parts of some result cannot be attributed individually to different Parties), it will be jointly owned, unless those Parties concerned agree on a different solution. The signed CA includes more specific rules regarding the exploitation (including the right to apply for patents) of jointly developed Results.
- These rules will in no way contradict the provisions on the exploitation of Results and Joint Ownership established by the Grant Agreement.

Industrial property rights assets

PROLIFIC proposes a complete range of activities leading to the optimal visibility of the project and its results finally materialised in the form of at least 16 product prototypes, increase the likelihood of market uptake. Knowledge and know-how together with the support of an efficient IPR manager will ensure a smooth handling of the individual intellectual property rights of the involved partners in view of paving the way to knowledge transfer.

The overall IPR strategy of the project is to ensure that partners are free to benefit from complementarities and to fully exploit their market position. Regarding ownership of the foreground, it is agreed that foreground will be the property of the beneficiary carrying out the work generating that foreground. Where several beneficiaries have jointly carried out work generating foreground and where their respective share of the work cannot be ascertained, they will have joint ownership of such foreground. They will establish an agreement regarding the allocation and terms of exercising the joint ownership, including definition of the conditions (compensation to other joint owner(s), time plans *etc.*) for granting licenses to third parties. Conditions of the transfer of ownership of the own foreground to 3rd parties has been predefined in the signed CA ensuring that the rights of the other project Partners will not be affected by such transfer. Beneficiaries will also define the background needed for the purposes of the project development in a written agreement and, where appropriate, may agree to exclude specific background. Access rights to foreground and background will be granted to the other beneficiaries, if it is needed to enable those beneficiaries to carry out their own work under the project and/or to use their own foreground provided that the beneficiary concerned is entitled to grant them. Such access rights shall be free of any administrative transfer costs and will be granted under fair and reasonable conditions or on a royalty-free basis unless otherwise agreed by all beneficiaries in a specific agreement. All access rights will be granted upon written request.

Prototype physical results, or that arise from the activities in this project will be owned by all parties that have provided some value to it. Each part is individually owned the physical part that contributed. With regard to scientific publications relating to foreground published before or after the final report details/ references and an abstract of the publication must be provided to the Commission at the latest two months following publication. Furthermore, an electronic copy of the published version or the final manuscript accepted for publication shall also be provided to the

Commission at the same time for the purpose set out in Article II.12(2) if this does not infringe any rights of third parties. The Parties undertake to cooperate to allow the timely submission, examination, publication and defence of any dissertation or thesis for a degree which includes their Foreground or Background.

The concerned project partners receive access rights to the project results under reasonable conditions. For use of project results in research and teaching without any commercial character the respective project partners receive access rights on royalty free basis. PROLIFIC envisaged key knowledge items are:

- Selection of bioactive extracted from co-streams and ranking according to the suitability of biochemical composition and performance as to the defined final applications and products.
- Innovative multistep approaches to extract the selected compounds from agri-food residues.
- Optimization of the biorefinery concepts to keep the functionality of the components ensuring the cost-effectiveness of the processes.
- Validation of nutritional, antimicrobial, antibiotic, toxicological and rheological properties of the active principles and quantification of the advantages compared with existing commercial available formulations.
- New high added value products for food, feed, packaging material and cosmetics including the new bio-compounds obtained from coffee, legume and fungi residues.

In the preparation of this proposal the complementarity of the members of the consortium was carefully analysed in order to avoid and prevent as much as possible potential conflict of interests. Besides, during the preparation of the project a careful literature survey was performed, and no limitation for the freedom to operate of the project was identified.

d) Measures to facilitate market exploitation

Beside the fact that technologies to recover compounds and fractions will be developed with and for industrial end-users producing the prototypes, further measures will be adopted and implemented during the PROLIFIC project in order to ensure the success of future market exploitation of the results. Competitiveness of high performance products will always include a “cost effective” vision in the development of every stage of the value chains.

During PROLIFIC execution: The R&D&I efforts will be focused on the efficient use of resources and efficient processes to ensure future exploitation success:

- WP1: Inventory and characterisation of the streams: Full inventory and ranking of the active compounds to be extracted will be used as competitiveness factors for end products development.
- WP2, WP3, WP4: Extraction, purification, modification, formulation. Advanced technological processes will be deployed to optimize each step of the value chains from the cost-effectiveness point of view.
- WP5: Validation of final application products (incl. consumer satisfaction questionnaires). Production processes for final product prototypes will be adjusted in order to enable the manufacturing of competitive final products.
- WP7: Intensified relation with business angels to prepare fund raising for product development needing additional financial support

Preliminary Business plans

Preliminary business plans will be drafted for the different ingredients and prototype products (WP7, Task 7.2) in order to foster relevant partners to commercially exploit the results of their innovation efforts and improve their competitiveness in their respective sector. The preliminary business plans will include an analysis of the external and internal environment as well as the necessary financial, operational and risk analyses including sustainability plan for the future commercial viability of the technical solution. These business plans canvas and more advanced drafts will constitute strong basis for pitching in front of business angels in the frame of event(s) dedicated to fund raising.

Table 8. Foreseen path to draft business plan canvas for PROLIFIC technologies, ingredients and prototypes

Area	Technologies and PROLIFIC results	Potential customers	Partner responsible	Exploitation mean and strategy
Research and Development	Extraction, purification methods	Extraction and Purification companies	UNIPR	Joint protection of the invention by trade secret contract or patent.
	Downstream processing	Industrial end Users	BBEPP	
	New methods for fraction conversion	Extraction and Purification companies	UNIPR/ FHNW	
Downstream and upstream	Formulation	Extraction and Purification companies	IGV, SSCIA, COSM, SB, NS, FTEN	Trade secret contracts with RD centers or commercial patents.
Extraction and formulation	Food, cosmetics, feed packaging end users	Industrial end Users	IGV, COSM, SB, SSICA, NS, FTEN	Trade secret contracts with RD centers or commercial patents.
Validation and safety/toxicity tests	Validation technologies	Industrial end Users	CELAB	Trade secret contracts with end-user associations.
Functional end products	Industrial prototypes	General Public, Livestock farming	IGV, NS, COSM, FTEN, SB, SSICA	New products to the market with new properties.

As previously explained, the exploitation of the results within this project is warranted by the presence of relevant industrial partners in the whole value chain, which have a relevant positioning in national and international markets, ensuring realistic market penetration by the products. We have performed a preliminary market analysis either at EU or regional market level as well as at partner company for some examples of the produced ingredients or of product prototypes 3). This analysis includes the opportunities and needs for product exploitation and concludes with estimations on foreseen percentage of increase of volume of sales in the following years after project execution.

Table 9. PROLIFIC preliminary market analysis of examples of ingredients and product prototypes

Sector	Product prototype	EU market/ Company	Current sales volume (million €)	Foreseen increase of sales volume
	Ingredients/additives			
Food	Proteins/peptides	World global market	25000	10%
Food	Proteins/peptides vegetal origin	EU market	2000	10%
	Prototypes			
Food	Vegan and vegetarian markets	IGV	1.25	25%
Food	Fermented and cooked sausages	SSICA (Italian market)	1724	5%
Packaging	Biodegradable food and cosmetic packaging[93]	EU market	1655	21% in 8 years
Packaging	Biodegradable food packaging	EU market	667	21% in 8 years

After the project: Sales and marketing campaigns will aim at promoting the properties and healthy uses of the compounds and fractions as valuable ingredients in food, feed, cosmetics and packaging applications. We will also promote the more sustainable use of agri-food residues as natural source among different stakeholders as policy makers, end users associations, *etc*. at regional, national and European level.

[93] https://www.coherentmarketinsights.com/market-insight/biodegradable-packaging-market-395

All measures will aim at facilitating the acceptation and introduction of bio-components from agri-food residues origin into food, feed, cosmetics and biopolymer sectors, enhancing their properties and demonstrating the advantages of the use of new natural resources for more sustainable and improved applications and products.

e) Regulatory and/or standard compliance

All partners from PROLIFIC consortium are committed to fully comply with all applicable regulations and other legal requirements to conduct their research in a highly ethical and legal manner.

Table 10. Actual regulations related to PROLIFIC activities

Regulation	Title / content
	Food industry
Regulation 1924/2006/EC	EC Regulation on Nutrition and Health claims made on foods
Regulation 1830/2003/EC	EC Regulation on traceability and labelling of GMOs and of food and feed products produced from GMOs
Regulation 178/2002/EC	EC Regulation on Food Safety
Regulation 2283/2015/EC	EC Regulation on Novel Foods and Novel Food Ingredients
Regulation 1829/2003/EC	EC Regulation on Genetically Modified Food and Feed
Regulation 1333/2008/EC	EC Regulation on Food Additives
Regulation (EU) 1169/2011	EC Regulation on the provision of food information to consumers
Directive 2009/32/EC	EC Directive on extraction solvents used in the production of foodstuffs and food ingredients
	Cosmetic industry
Regulation 1223/2009/EC	It strengthens the safety of cosmetic products and streamlines the framework for all operators in the sector. In particular: – GMP (Good Manufacturing Practices) Certification – Cosmetics Testing and Cosmetic Safety Assessments – Cosmetics Ingredient Profiles and Product Information Files (PIFs) – Cosmetic Product Safety Reports (CPSR) – Part A Cosmetic Product Safety Information – Part B Cosmetic Product Safety Assessment – Regulatory Compliance Support and Regulatory consulting notifications – Clinical Studies for Claim Substantiation – Design, placement, and monitoring and Product Labelling Review and Support
	Packaging
Regulation 1935/2004/EC	EC Regulation on materials and articles intended to come into contact with food
Regulation 2023/200/EC	EC Regulation on good manufacturing practice for materials and articles intended to come into contact with food
Regulation 10/2011/EC	EC Regulation on plastic materials and articles intended to come into contact with food
Regulation 450/2009/EC	EC Regulation on active and intelligent materials and articles intended to come into contact with food
	Feed industry
Regulation 1831/2003	Authorization, supervision and labelling of feed additives
Regulation 767/2009	Marketing and labelling of feed materials and compound feed
Directive 90/167/EEC	Mixing veterinary medicine into feed, its marketing and use
Directive 2002/32/EC	Contaminant and non-desirable substances in animal feed
Regulation 178/2002	Traceability (General Food Law Regulation)
Regulation 1830/2003/EC	EC Regulation on traceability and labelling of GMOs and of food and feed products
Regulation 1829/2003/EC	EC Regulation on Genetically Modified Food and Feed
Regulation 183/2005/EC	EC Regulation on requirements for feed hygiene
	Others
Regulation 14044	Environmental Management
UNEP specifications	Guidelines for Social Life Cycle Assessment of Products

2.3 Research data management

The project will use primary data collected during the project, and a tentative detailed description on how the different types of collected and generated data are handled by PROLIFIC project is reported below. Data will be formalised in structured databases for the purposes of elaborations to be carried out in the project. They will be used for achieving project objectives and related scientific dissemination activities, including follow up scientific activities. Data will primarily be used by the partner collecting and generating the data. Specific agreements to use data jointly involving partners that have not participated in data collection will be regulated during the project in relation to specific authorship objectives and not involving monetary transactions. When the same methodology for data collection is used, pooled database will be built to maximize the potential exploitation of the effort exerted during the project. In this case each partner can continue to use independently the data that it has collected, while the use of pooled data is granted based on authorship agreements (for scientific publications) or project roles (for deliverables). In case secondary data are used, they will be requested and managed by the partners in charge of the related task and using according to the rules of the providers, including non-delivery to other project partners if this is not part of the agreement taken. Data may be used in collaboration with individuals affiliated to non-partner institution when this may be useful to maximise impact and do not infringe the Knowledge and Innovation Management strategy of the PROLIFIC Consortium. In this case, data use will be granted under a specific non-disclosure agreement. Ethical issues will be handled during the Project lifetime, according to the signed versions of the Grant agreement and the Consortium agreement. PROLIFIC Project recognises the value of regulating research data management issues, and foresees the delivery of a Data Management Plan (DMP) at Month 6 as relevant Deliverable in WP7. The DMP will define how research data will be handled during the life time of the PROLIFIC project, and after it is completed. It will be a dynamic document that will be delivered under the leadership of the Project Coordinator and with a strong involvement of the Steering Committee. All partners will actively participate not only in the draft of the DMP at Month 6, but in its dynamic development over the time, consisting in its continuous update according to possible needs, evolving expectations and framework conditions. Status of the DMP will be discussed at GA meetings, and according to the Guidelines for Data Management of Horizon 2020, a more elaborated version of the DMP will be released at Month 42, when the PROLIFIC Consortium will have the most precise understanding of the progression of the exploitation plan as well as of the dissemination and communication plan.

Types of data Prolific will collect and generate

- Data about feedstocks (composition, volumes, seasonality, etc.)
- Data on the regulations for the targeted sectors
- Data on technical solutions to valorize the feedstocks
- Data on the formulation of prototypes
- Data on the validation of the prototypes and consumer acceptance
- Data used to draft business plans
- Set of LCA and CEA data

Standards to be used

- Standards that can help include management standards to optimize processes, such as ISO 9001 (quality), ISO 14001 (environment) and ISO 50001 (energy).
- Standards that can be used to communicate environmental impact to inform consumer choice: ISO 14020, ISO 14021, ISO 14024 and ISO 14025.
- And all regulatory documents listed in Table 10

Way to exploit and/or share/make accessible for verification and re-use

Every WP leader is responsible for the storage and the compliance of data and then upload to INTRANET of PROLIFIC website. With the aim to encourage data owners and curators of other existing databases of local, national, European authorities, networks and policy boards to collaborate in sharing information, a **clear and detailed process for IPR** will be developed. DMP ensure data **long-term maintenance**. WP7 will develop a strategy to encourage i) the identification of the most-suitable data sharing and preservation methods; ii) the feeding of the database with reliable data (those following the guidelines for reporting the minimum information) and iii) its use via user-friendly interface, clear rules on its accessibility and an efficient quality assurance and quality control of the submitted data. The post-project maintenance, exploitation and management of the PROLIFIC results will be completed coherently combining the needs, opportunities and specifications that will be arise in the DMP and the Exploitation Plan design.

<u>Way to curate and preserve the data</u>

PROLIFIC will make all efforts for ensuring its data long-term maintenance. During the DMP co-development, it will be highlighted a strategy to encourage *i)* the feeding of the intranet with reliable data (those following the guidelines for reporting the minimum of information) and *ii)* its use via a user-friendly interface, clear rules on its